



Process Flow & P&ID's

(Process Engineering Drawings)

Part of the
Petroleum Processing Technology Series



THE OIL & GAS ACADEMY

Process Flow & P&IDs

Process Engineering Drawings

(Part of the Petroleum Processing Technology Series)

Contents

Page

BOOK 2

* Section 1 - Symbols

3

- 1.1 Structure
- 1.2 Pipeline Symbols
- 1.3 Other Lines
- 1.4 Pipeline Numbering and Identification
- 1.5 Product Designations
- 1.6 System Numbering
- 1.7 Pipeline Numbering
- 1.8 Pipeline Specifications
- 1.9 Insulation
- 1.10 Pipe Fittings
- 1.11 Valves
- 1.12 Valves used for "On/Off" Service
- 1.13 Valves used for "Control" Service
- 1.14 Valves used for "One-Way" Service
- 1.15 Valves used for "Special" Duties
- 1.16 Valve Actuators
- 1.17 Tanks and Pressure Vessels
- 1.18 Tank and Vessel Fixtures and Fittings
- 1.19 Filters
- 1.20 Pumps and Compressors
- 1.21 Metering Devices
- 1.22 Heat Exchangers
- 1.23 Other items of Equipment
- 1.24 Equipment Identification
- 1.25 Instrument

* Section 2 - Practical Application of Symbols

40

- 2.1 Plate Type Heat Exchanger
- 2.2 Valve Interlocks

* Section 3 - Piping and Instrument Diagrams

44

Figures 7 to 17 to be used with Book 1 Sections 2, 3 and 4

Section 1 - SYMBOLS

1.1 STRUCTURE

In this Section we will look at the different symbols which may be used on Piping and Instrument Diagrams (P&IDs) (also called Process Engineering Drawings). Although BS1553 provides the specifications for symbols, you will find many variations in the different P&IDs you will come across. The symbols given in this appendix include the BS1553 specifications, and some of the most common variations of symbols used in the oil and gas industry.

The Appendix has been broken down into different categories. They are :

- pipeline symbols
- pipeline numbering and identification
- pipe fittings
- valves
- valve actuators
- tanks, separators and common pressure vessels
- filters
- pumps and compressors
- metering devices
- heat exchangers
- other items of equipment
- equipment and instrument identification

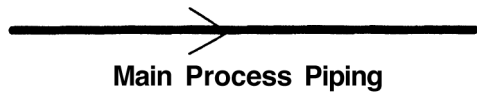
In each section there is an example of the British Standard symbol, an example of any common variation of the symbol, and a brief description of the main points of the item.

After studying this Appendix, and applying the knowledge learned in the POL Unit, you should be able to **navigate** your way through any P&ID.

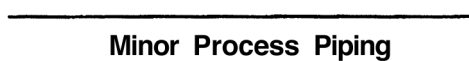
It should be appreciated that P&IDs are **NOT** scale drawings. However, the actual pattern of pipeline connections, pipe fittings, valves, instruments etc **WILL** be accurate.

1.2 PIPELINE SYMBOLS

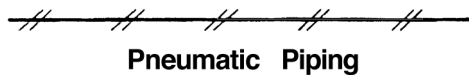
In general, the symbols used to identify the different types of pipe will be as laid out below :



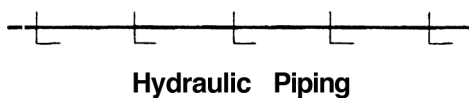
Usually includes all piping above 2" diameter. In the diagram the flow is indicated as being from left to right.



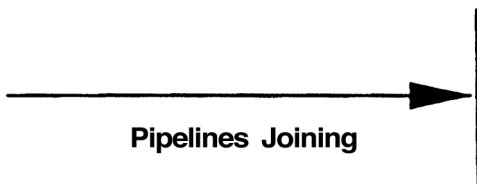
Usually includes all piping less than 2" diameter and all utility piping. The direction of flow is not normally indicated.



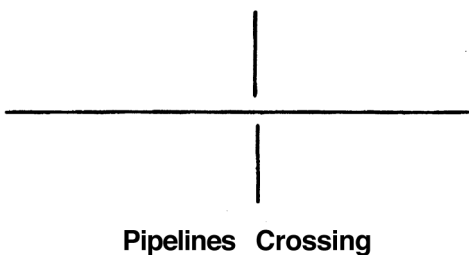
Most control valves are pneumatic (air), powered by Instrument Air.



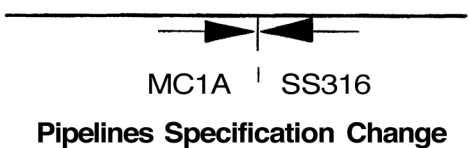
Most wellhead valves and pipeline Emergency Shutdown (ESD) Valves are hydraulically powered. The hydraulic fluid may be water or oil based.



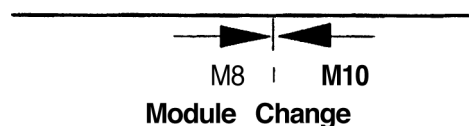
Flow from the horizontal pipe joins the flow in the vertical pipe.



The two pipes are crossing each other on the drawing.



At the indicated point the pipeline material specification changes.



At the indicated point the pipeline leaves one Module or Plant Area (M8) and enters another Module or Plant Area (M10).

1.3 OTHER LINES

Electronic Signal

A low voltage electric signal. Usually to and from controllers or instrument switches.

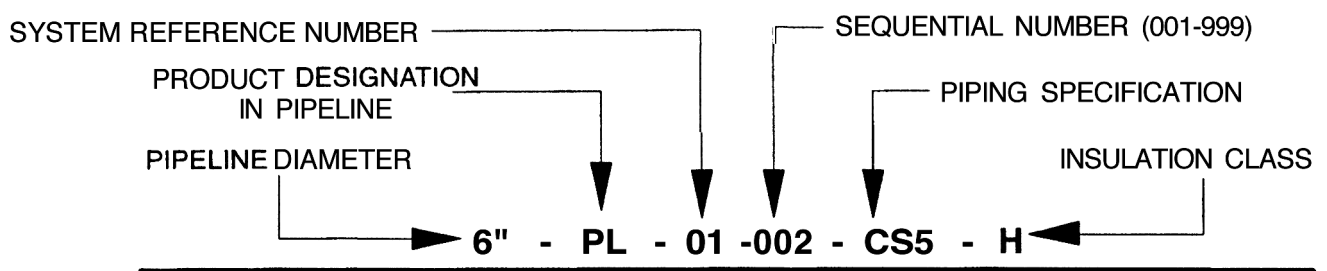
—○—○—○—○—○—
Software Link

A signal to and from computer based instruments.

1.4 PIPELINE NUMBERING AND IDENTIFICATION

All pipelines will have a unique identification number. As a general rule the identification number will indicate :

- the pipe diameter Note : ID size given up to 12" dia. OD size given for over 12"dia.
- the product which is carried within the pipeline
- the system number in which the pipeline is installed
- the pipeline sequential number which identifies the particular pipeline within each system
- the pipeline specification (ie the pipeline pressure rating and the material from which the pipeline is made)
- the type of insulation applied to the pipeline



Illustrates a 6", carbon steel, process liquids pipeline, insulated for full heat conservation.

1.5 PRODUCT DESIGNATIONS

A few of the more common pipeline Product Designations are listed below, with a few alternative types of identification: -

- PL = Process Liquid may also be HC = Hydrocarbons, PO = Produced Oil
- PG Process Gas
- FG = Fuel Gas ; -
- AI = Instrument Air, IA sometimes used
- AP = Plant Air, PA sometimes used
- CW = Cooling Water, may also be CM = Cooling Medium, SW = Seawater

Figure 17 provides a comprehensive list of product designation codes.

1.6 SYSTEM NUMBERING

The different systems within each process are normally identified by a number, eg The Wellheads and Manifolds System may be System 01. The Crude Oil Separation System may be System 02. The Produced Water Treatment System may be System 12. The Instrument Air System may be System 62. An instrument air line which passes through the crude oil separation area and the produced water area, will still be designated a **System 62** line.

1.7 PIPELINE NUMBERING

A different number is usually given for each length of pipeline. The number will usually change when something happens to change the nature of the pipeline, eg When the pipeline specification changes across an emergency shutdown valve, when the diameter of the pipeline changes across a reducer etc.

1.8 PIPELINE SPECIFICATIONS

The method of identifying the pipeline specification can be extremely complex, due to the large number of pipeline materials and pressure ratings. A common system is where letters indicate the pipeline material and a specific number indicates the pressure rating. A few examples are :

- CS1 = Carbon Steel - Schedule 40
- SS2 = Stainless Steel - Schedule 80
- CU1 = Copper - Schedule 40

1.9 INSULATION

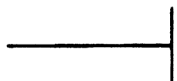
Pipeline insulation, when provided, will mainly be for:

- H = Full Heat Conservation
- HE = Full Heat Conservation with electrical trace heating
- Z = Cold Conservation
- P = Personnel Protection
- F = Frost Protection
- FE = Frost Protection with electrical trace heating
- A = Acoustic Protection
- R = Fire Proofing

Combinations of insulation class may also be used. A pipeline designated as insulation class P/A would indicate that the pipeline required personnel protection insulation **and** acoustic protection.

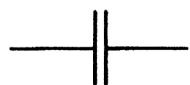
1.10 PIPE FITTINGS

Any item of equipment which can be attached or connected into a pipeline may be classed as a **pipe fitting**. The majority of the examples given below will be found somewhere on most oil and gas production and treatment plants. A few specialised items have also been included.



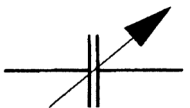
Flange

A flange is a fitting which is welded or screwed on to the end of the pipe. The flange allows the pipe to be joined up to another pipe, a pipe fitting or item of equipment. If an item of equipment is shown without a flange attachment it most often indicates that the item is welded or screwed into the pipe.



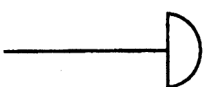
Two Flanges

The flanges are joining two pipes. A gasket placed between the two flanges ensures that the joint is sealed. Most pipe fittings and items of equipment are fitted into the pipeline between two flanges.



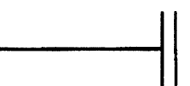
Insulated Flanges

Insulating gaskets, bolt sleeves and washers are installed to insulate one flange from the other. Fitted to sections of pipeline which are protected from corrosion by impressed current cathodic protection systems, or where a difference in metallurgy could start the corrosion process, eg Carbon steel pipe / bronze valve.



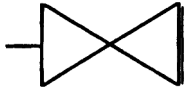
Pipe Cap

Used to terminate a pipe in situations where there are no plans to extend or fit anything to the end of the pipe in the foreseeable future.



Blank Flange

Used to terminate a pipe in situations where there may be a reason to extend or fit something to the end of the pipe in the future.



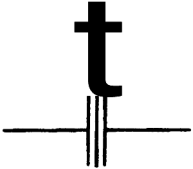
**Blank Flange
with Valve**

Used to terminate a pipe in situations where there may be a reason to extend or fit something to the end of the pipe in the future, but in situations where it may not be possible to depressurise the pipe or take it out of service.



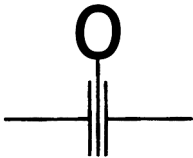
Hose Connection

Used to terminate a pipe in situations where there will be a regular requirement to connect a hose to the pipe.



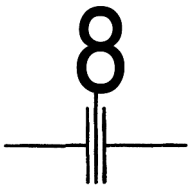
Pipe Spade / Blind

Flat solid plate. Fitted in locations where a positive isolation is required to prevent flow through the pipe.



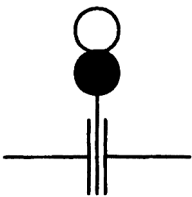
Ring Spacer

Flat plate with central hole. Fitted in locations where the installation of a pipe spade blind may be required.



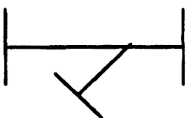
Spectacle Blind

A combined pipe blind and ring spacer. Fitted where frequent positive isolation may be required. The example shows that the blind is normally in the open position.



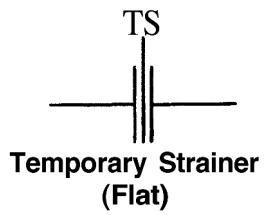
Spectacle Blind

The example shows that the blind is normally in the closed position.

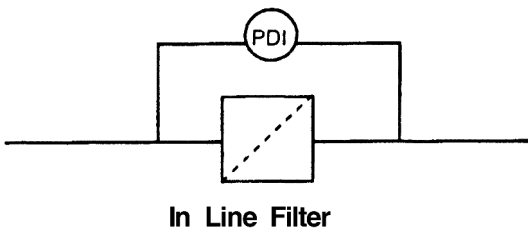


"Y" Strainer

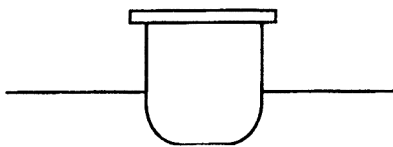
A Filter. Fitted where fine screening is required but where frequent changes are not expected.



Coarse filter. Fitted to protect equipment from construction debris (eg gloves, welding rods etc).

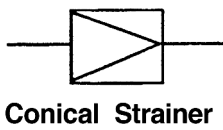


Basic symbol for filters / strainers that are usually fitted with mesh baskets or cartridge type filters. Sometimes provided with pressure differential indicator (PDI).

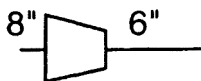


Alternative symbol.

Filter / Strainer



Coarse Filter. Also called a witches hat. Fitted to protect equipment from construction debris (eg gloves, welding rods etc).



Fitted where a pipeline size change is required.

Reducer (8" to 6")

1.11 VALVES

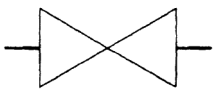
Before looking at the symbols used to identify the different types of valve we will consider the different services which the valve may be required to perform. The three main service requirements are :

- **ON/OFF SERVICE** : For on/off service the valve should ensure full flow when fully open and a leak free shut-off when fully closed.
- **CONTROL SERVICE** : In control service the valve should be able to control the flow of fluid through the valve in accordance with the requirements of the design. The valve should also be able to give a leak free shut-off when it is fully closed.
- **ONE WAY SERVICE** : Valves are required which ensure that flow is maintained in only one direction. They should allow free flow in the direction required but give a leak free shut-off in the reverse direction.

We can see from the various requirements of each service that we will require different types of valve. I will explain the basic design features of the different types of valve.

On most P&IDs each type of valve is given a different symbol. The type of valve selected will mainly depend on the operating conditions, product and type of service. Other factors such as cost, weight and maintenance requirements will also be considered.

In some cases the different types of valve may not be indicated on the P&IDs. When this occurs, generic valve symbols as shown below are sometimes used.



Valves which are normally in the **OPEN** position will not usually be coloured in. The letters “NO” (indicating **normally open**) may also be printed next to the valve.



Valves which are normally in the CLOSED position will usually be coloured in. The letters “NC” (indicating **normally closed**) may also be printed next to the valve.

1.12 VALVES USED FOR “ON/OFF” SERVICE

The following valves are the ones normally selected for **ON/OFF** service. They generally allow a free flow when the valve is fully open and a leak free shut-off when the valve is fully closed.



Wedge Gate Valve

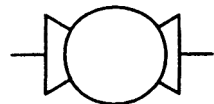
Most common of all valves. Used at all pressures. Gate valves are not very good in dirty service as debris may damage sealing surfaces or accumulate at the bottom of the valve to prevent closure. The gate valve must **never** be used for control service as the flow across the valve will cut away the sealing surfaces.



Wedge Gate Valve

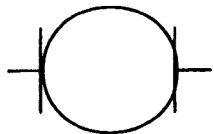
Note : The gate valve symbol may be used as a common symbol for all valve types.

Alternative symbol.



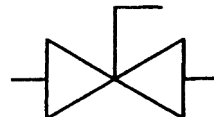
Ball Valve

Used at all pressures. Some designs have sealant injection points to improve shut-off capability. Not very good in dirty service because debris may damage seals. Specialised internal designs may allow the valve to be used for flow control with relatively low pressure drops.



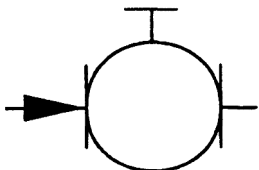
Ball Valve

Alternative Symbol.



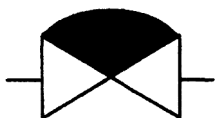
Plug Valve

Mainly used in medium and low pressure service. Most designs have sealant injection points to improve shut-off capability. Not very good in dirty service because debris may damage seals.



Orbit Valve

Mainly used in specialised high pressure service. A derivative of the ball valve which has a **turning / sliding** action which pushes the ball against the sealing surface. Must be facing correct way in line for best results. Performs fairly well in dirty service.

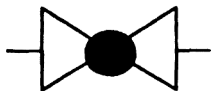


Diaphragm Valve

Used mainly in low pressure dirty services. Care should be taken not to over tighten the valve and damage the flexible diaphragm.

1.13 VALVES USED FOR “CONTROL” SERVICES

The following valves are able to control the flow of fluid through the valve and, in most cases, give a leak free shut-off when fully closed.



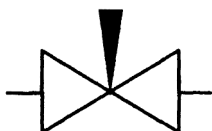
Globe Valve

Used at all pressures. The most common of all **control service** valves. Good in dirty service. Different internal designs can cope with all service and pressure requirements.



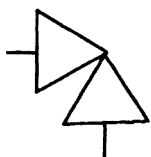
Needle Valve

Used at all pressures. A derivative of the globe valve. Used for very fine flow control (eg sample points). Useless in dirty service.



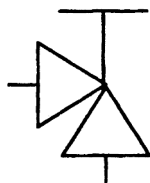
Needle Valve

Alternative symbol.



Angle Valve

Mainly used at high pressures. Derivative of the globe valve. Reduced turbulence within the valve gives better flow than the globe valve.



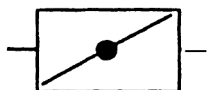
Choke Valve

Used for high pressure drop service. A derivative of the angle valve.



Butterfly Valve

Mainly used in low pressure and low pressure drop services. Some designs are directional in order to improve sealing. Not to be relied upon for tight shut-off.



Butterfly Valve

Alternative symbol.

1.14 VALVES USED FOR “ONE-WAY” SERVICE

Valves used for **ONE-WAY** service are called **CHECK VALVES**, **NON-RETURN VALVES** or **ONE-WAY VALVES**. The following valves ensure that flow is maintained in only one direction and, in most cases, allow free flow in the direction required but give a leak free shut-off in the reverse direction. The three main versions are :

- **SWING CHECK VALVE** : A flat circular plate is hinged so that it lifts to allow flow past the plate in one direction but falls down to seal against the valve seat when the flow is reversed. The Swing Check Valve is the most common type of valve in **ONE-WAY** service.
- **TILTING PLATE CHECK VALVE** : A flat circular plate is hinged with a slight offset from the central position. The offset position of the hinge results in the valve opening in one direction but sealing in the opposite direction. Mainly used in high pressure, high flow gas service.
- **BALL CHECK VALVE** : A free moving ball is contained within a cage. The ball lifts away from the seat to allow forward flow but falls back into the seat when the flow is reversed. Mainly used in low flow liquid service.
- **PISTON CHECK VALVE** : A free moving piston slides up and down inside a cage. The piston lifts away from the seat to allow forward flow but falls back into the seat when the flow is reversed. Mainly used in low flow, high pressure liquid service.

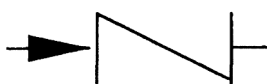
Any of the above types may be spring loaded to assist the sealing function when the flow is reversed. Swing check valves may also be fitted with :

- a device which allows the check valve to be **screwed down** to enhance the tight shut off capabilities,
- a hydraulic dampener which prevents **slam shut** closure.



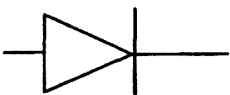
Check Valve

Indicated flow is from left to right.



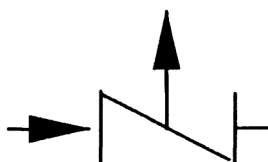
Check Valve

Alternative symbol. Indicated flow is from left to right.



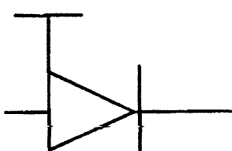
Check Valve

Alternative symbol. Indicated flow is from left to right.



Check Valve with
Recycle Facility

Main flow is indicated from left to right. Recycle flow is indicated vertically. Used to provide centrifugal pumps with a discharge check valve which also incorporates a **minimum flow** facility.

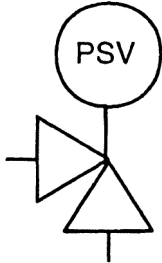


Check Valve with
"Screwed Down" Facility

Check valve may be **screwed down** to enhance tight shut-off. Occasionally found on high pressure pumps or compressors which share common headers with other pumps or compressors. Indicated flow is from left to right.

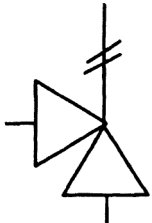
1.15 VALVES USED FOR “SPECIAL” DUTIES

The following valves are designed to cope with certain special requirements. They are all specialised derivatives of valves which have already been described.



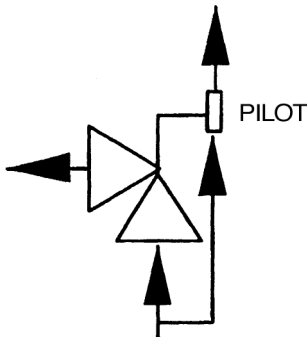
Used to protect vessels and pipes from over pressure. Derivative of the angle valve. Standard type is spring loaded to ensure that valve lifts at pre-set pressure underneath the valve seat. **Balanced** models are available which compensate for any difference in pressure downstream of the valve.

Pressure Safety Valve
(Also called a Pressure Relief Valve)



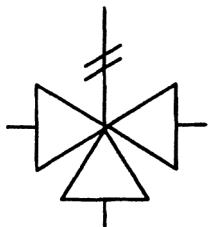
Alternative symbol.

Pressure Safety Valve



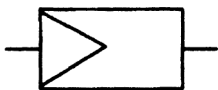
Used to protect vessels and pipes from over pressure in high pressure high volume services. A small spring loaded pressure relief valve (the **Pilot**) activates to allow the main valve to open.

**Pilot Operated
Pressure Safety Valve**



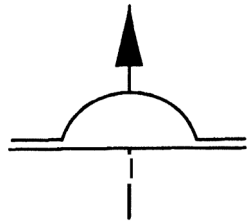
Used to protect tanks and low pressure vessels from over pressure and vacuum conditions. Allows air to move into and out of the tank or vessel in response to changing internal pressure. Derivative of the globe valve which uses weighted valve seats.

Pressure Vacuum Valve



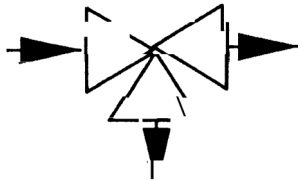
Excess Flow Valve

Used to prevent excess flow to or from tanks and vessels. Usually positioned where piping failures could occur which may have extremely hazardous consequences (eg flexible hoses carrying hydrocarbons).



Rupture Disc
(Also called a "Bursting Disc")

Not strictly a **valve** but used for pressure relief service in a similar manner to a pressure safety valve. Often positioned beneath PSV's to protect them from corrosive process fluids. The rupture disc is also used as a **last resort** over pressure protection device in critical services, such as the shell side of a shell / tube heat exchanger.

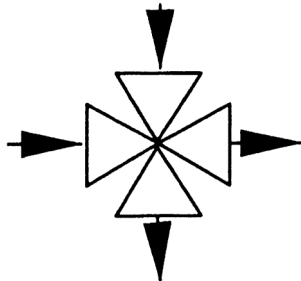


Three-Way Valve

A single inlet is split into two outlets. The internal design may be :

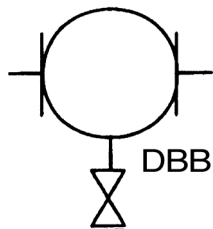
ON/OFF - depending upon the valve position, the flow may be through either outlet.

SHARED - depending upon the valve position the flow may be through either or both outlets.



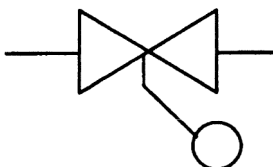
Four-Way Valve

Two inlets may be diverted to two outlets. Specialised valve used on meter prover loops.



Double Block and Bleed Valves

May be derivatives of ball valves (as illustrated) gate valve or plug valves. An internal section of the valve is sandwiched between two independent seals. The internal section is connected to a **bleed** valve. The bleed valve may be opened to drain the internal section, and prove that there is no migration of fluid across the two sealing surfaces. Double block and bleed valves are most commonly found on metering systems.



Float Operated Valve

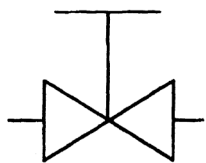
Used for maintaining levels in storage tanks such as, potable water, diesel etc.

1.16 VALVE ACTUATORS

Valve Actuators are the devices which move valves to the desired position. As the majority of valves are **hand actuated** the most common valve actuator is the Process Operator! Other common actuators which may be found are :

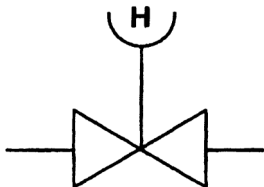
- **DIAPHRAGM ACTUATORS** : A flexible diaphragm is moved in and out by pneumatic or hydraulic pressure. Almost all **control** valves are pneumatically powered diaphragm actuated valves.
- **PISTON ACTUATORS** : A piston is moved in and out of a cylinder by pneumatic or hydraulic pressure. Almost all well safety valves and many sub-sea safety valves are hydraulically powered piston actuated valves.
- **MOTOR ACTUATORS** (Also called **MOTOR OPERATED VALVES**): An electric motor, suitably geared. Large valves in non-critical service are often fitted with a Motor Actuator.
- **SOLENOID ACTUATORS** : A solenoid is an electro magnetic device with limited movement. Almost all instrument air dump valves will be solenoid actuated.

The type of actuator will normally be indicated on the Piping and instrument Diagrams, but the type of valve being actuated may not be specified. The main valve actuator symbols are indicated below :



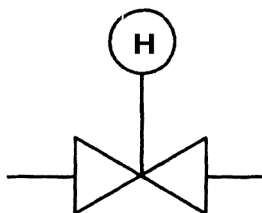
Hand Operated

If no symbol is attached to the valve it will also be a **hand operated** valve.



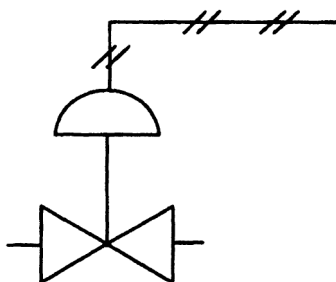
Manual Actuating Element

May be pneumatic, hydraulic or electric motor actuated by manual operation only.



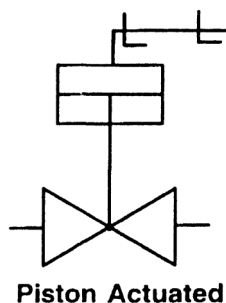
**Automatic Actuating Element
with Integral Manual Actuating Element**

Similar to above, but can be automatic operation as well.

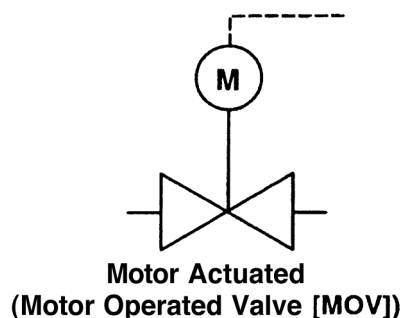


Diaphragm Actuated

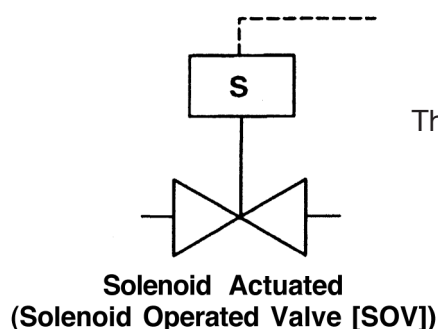
The illustration shows that the actuator is pneumatically powered.



The illustration shows that the actuator is hydraulically powered.

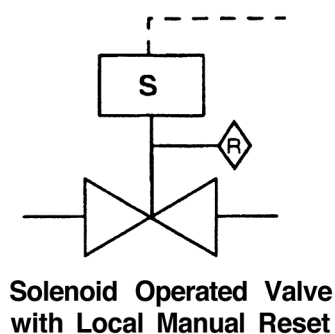


The illustration shows the electric power signal to the motor actuator.

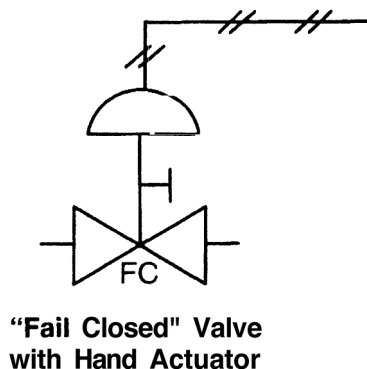


The illustration shows the electric power signal to the solenoid actuator.

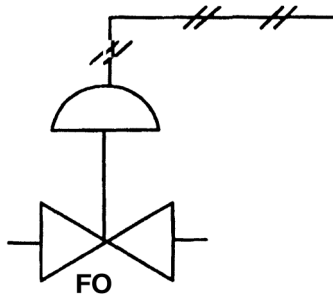
As well as identifying the type of actuator the Piping and Instrument Drawing will also indicate the main characteristics of the valve in the **FAILURE** mode :



SOV with a local reset facility to allow the signal to be reinstated to the valve.

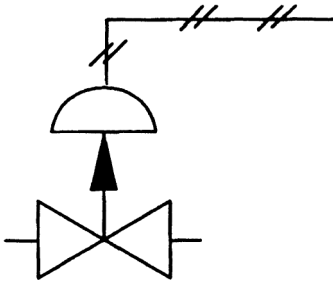


The valve can be opened by hand against the power being exerted by the closing spring if required. Occasionally a hand actuator is fitted with a clutch which can be used to both open and close the valve. **Beware of leaving these valves in the hand actuated position when normal operations are resumed** as they will not operate in response to the automatic signal.



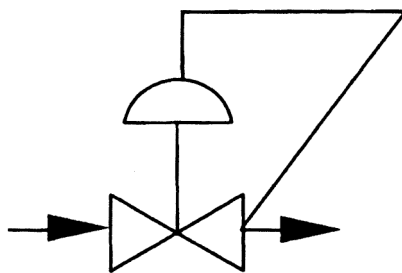
"Fail Open" Valve

The letters "FO" indicate that, in the event of a pneumatic power failure (ie an instrument air failure) the valve will move to the **OPEN** position.



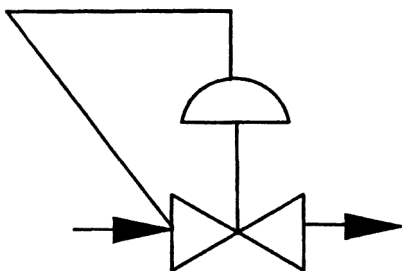
'Tail Open" Valve

An alternative signal. The arrow indicates that the valve will move to the **OPEN** position.



**Self Acting
Pressure Control Valve
with Downstream Tap**

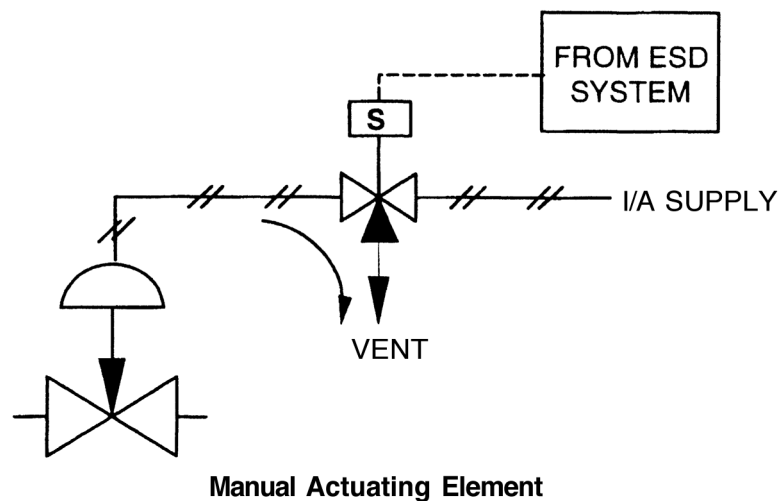
This PCV uses the flowing product to act on the diaphragm to control the downstream pressure requirements.



**Self Acting
Pressure Control Valve
with Upstream Tap**

Same as above but controlling upstream pressure requirements.

The configuration indicated below is typical for a diaphragm operated control valve in critical service.



Under normal operating conditions :

the ESD system ensures that there is a supply of electrical power to the solenoid operated valve

the solenoid valve is energised and in the **normal** position

the instrument air (I/A) supply is routed to the diaphragm valve through the solenoid operated three way valve

(The **normally closed** section of the three-way valve is shaded in to indicate that the normal flow is through the two open sections).

the control valve is in the **normal** position.

If the ESD system is activated :

the ESD system removes the electrical power to the solenoid operated valve

the solenoid valve is de-energised and moves to the **failure** position

the three-way valve changes position to :

- close off the supply of air from the instrument air system, and
- vent the instrument air from the diaphragm actuator

(The curved arrow shows the route the air takes when the solenoid valve is in the **failure** position),

the control valve moves to the **failure** position

(In the example, the control valve will fail to the **CLOSED** position, as indicated by the downward pointing arrow).

Study this system carefully. Ensure that you understand the relationship between the ESD System, the Solenoid Operating Valve and the Control Valve.

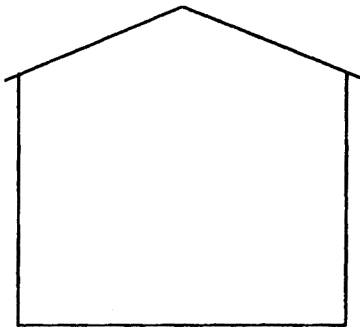
1.17 TANKS AND PRESSURE VESSELS

We will now look at a few of the tanks and pressure vessels used in the oil and gas industry. We will start by looking at the various types of tanks and then move on to pressure vessels and a few specialised vessels. The examples I have given cover most of the varieties of tanks and pressure vessels which may be encountered.



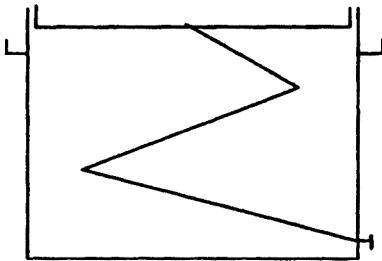
Open Tank

Seldom used in the oil and gas industry, and then only in water service. Open pits may be found on drilling rigs in mud service.



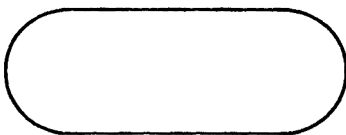
Cone Roofed Tank

Used offshore for the bulk storage of liquids. Used onshore for the bulk storage of low or non-volatile liquids. Cone roofed tanks in flammable liquid service are often gas blanketed. Often constructed with a **cone bottom** or a **sump** to allow complete emptying of the contents.



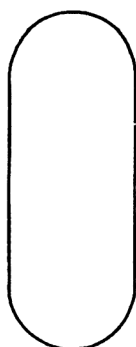
Floating Roof Tank

Seldom (if ever) used offshore. Used onshore for the bulk storage of volatile liquids. The roof floats on top of the product and reduces product losses by evaporation. (The first 1,000,000 barrel tanks ever constructed were floating roof tanks).



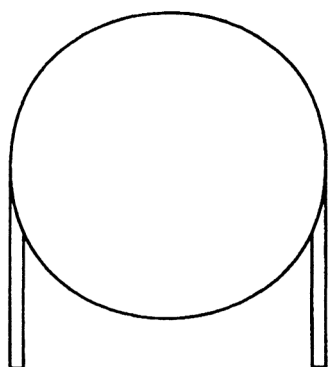
**Horizontal
Pressure Vessel**

Used in low, medium and high pressure storage or process services. As a general rule when the vessel is in process service the function will be part of the name, eg **separator**, **knock-out drum**, **surge drum** etc. When the vessel is in storage service it is generally called an **accumulator** or a **bullet**.



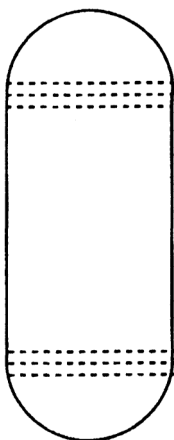
Vertical Pressure Vessel

Used in low, medium and high pressure process services, eg **knock-out drum, surge drum** etc. Very occasionally used in storage service.



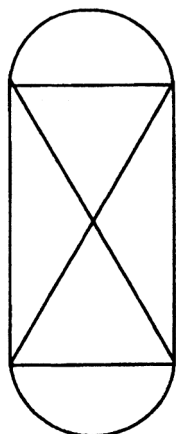
Spherical Pressure Vessel

Used for the storage of liquefied low vapour pressure gases (eg butane), sometimes referred to as a Norton Sphere. Not found offshore.



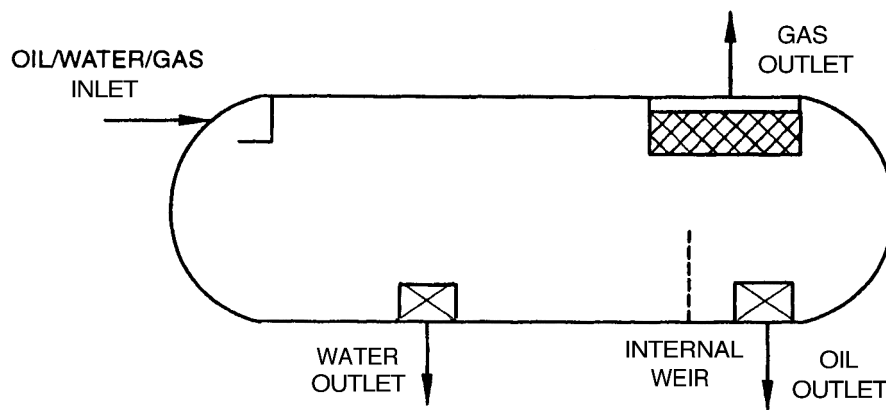
Trayed Distillation Column or Tower

A vertical pressure vessel used for all types of distillation, fractionation, rectification and stripping services. The most popular types of trays are **valve trays** but **sieve trays** and **bubble cap trays** may also be used. Used extensively offshore in water deaeration and glycol dehydration systems, and onshore in refineries and gas processing plants, for the separation of a wide range of hydrocarbon products.



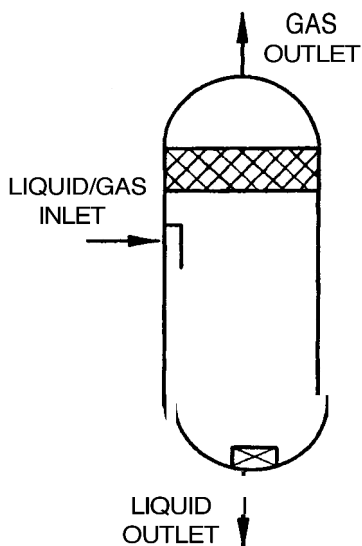
Packed Distillation Column or Tower

A vertical pressure vessel used for all types of distillation, fractionation, rectification and stripping services. The most popular types of packing are raschig rings, pall rings, ceramic or plastic balls and berl saddles. Used extensively offshore in water deaeration and glycol dehydration systems and onshore in refineries and gas processing plants for the separation of a wide range of hydrocarbon products.



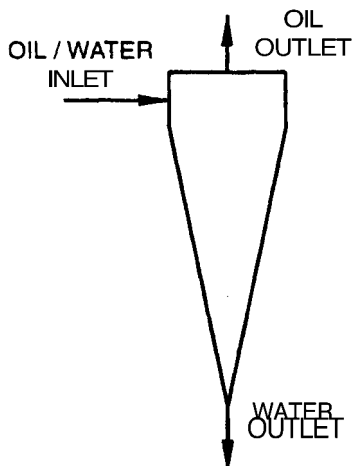
Horizontal Three Phase Separator

A horizontal pressure vessel which utilises gravity and a (relatively) long residence time (3 minutes) to separate gas and water from produced oil. The separated water is retained to the left of an internal weir. The separated oil flows over the internal weir to the oil outlet. The two liquid outlets are fitted with **vortex breakers** which prevents oil being drawn into the water outlet stream, and gas from being pulled into the oil stream. The separated gas leaves the top of the vessel after passing across a de-mister pad. The de-mister pad removes any entrained droplets of oil.



Vertical Two Phase Separator

A vertical pressure vessel which utilises gravity, centrifugal force and a reduced upward velocity to separate a small amount of liquid from a gas stream. The inlet is deflected around the inner walls of the vessel to create centrifugal force. The liquids strike the vessel wall and then drain down into the bottom of the vessel, where they are removed. The liquid outlet is fitted with a vortex breaker, which prevents gas from being pulled into the liquid stream. The gas leaves the top of the vessel after passing across a de-mister pad. The de-mister pad removes any entrained droplets of oil.



Cyclone Separator

A vertical, or horizontal, pressure vessel which uses centrifugal force. Cyclones may be used to separate :

- two immiscible liquids of different densities (eg oil/water separation)
- solids from a gas (eg dust extraction from a boiler flue)
- gas from a liquid (eg mud de-gasser)
- solids from a liquid

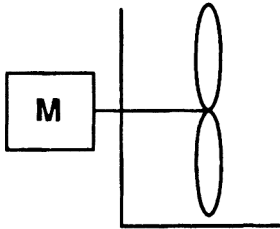
The oil/water cyclones used to separate oil from produced water are called **hydro-cyclones**. They are becoming more popular in produced water systems.

L18 TANK AND VESSEL FIXTURES AND FITTINGS



Manway / Access Hatch

All tanks and vessels are normally fitted with at least one **manway / access hatch** to provide access for personnel. Some are straightforward flanged connections, some are hinged, and others are provided with small cranes to make them easier to remove. As a general rule they are 24" in diameter and are fitted with an internal **grab handle** to make access easier.

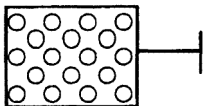


Propeller Mixer / Agitator

Tanks are often fitted with propeller mixers / agitators to ensure that the tank contents are kept mixed, or to keep any solids in suspension.

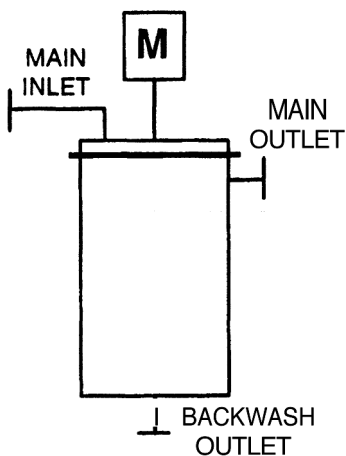
1.19 FILTERS

In the section on pipe fittings we saw four types of coarse filtering devices called **strainers**. These are also **filters**. I have classed them as **pipe fittings** because they are usually fitted as part of a pipeline, rather than as an individual item of equipment. Below we can see a number of filters which are fitted as items of process equipment.



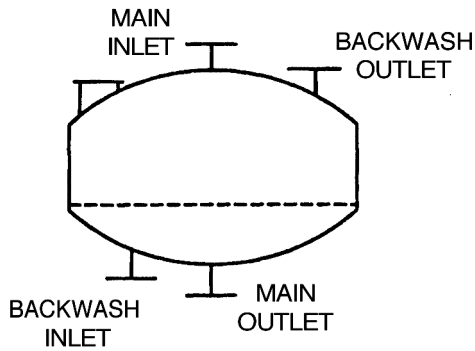
Strum Box

An extremely coarse filter which is used to prevent large items of debris entering a pipeline. Strum boxes may be fitted to the inside of dirty service tanks. They are most often found on the inlets to pumps in raw water service which take water from the sea or from a river.



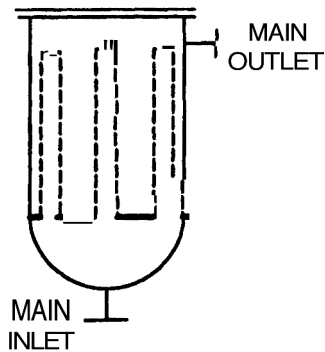
Basket Filter

Often found in raw water filtration service, downstream of the supply pump. Can be cleaned whilst still in service. A motor driven internal scraper revolves inside the basket and diverts filtered water back over the filter mesh to the backwash outlet. The cleaning cycle is usually activated by a timer or high differential pressure. The flanged top allows the filter basket and cleaning head to be maintained.



Sand Filter or Dual Media Filter

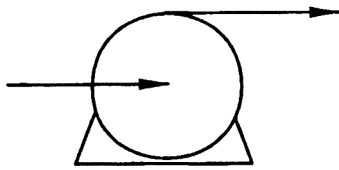
Extremely efficient method of filtration. Found in all types of service where fine filtration is required. Filter medium is graded layers of coarse and fine sand (sand filter) which may be overlaid with a layer of anthracite (dual media filter). The filter is cleaned by **backwashing** with filtered water. Some models have a **scouring** system which injects air or gas into the backwash stream to increase the cleaning efficiency. A manway allows access for the filter media to be changed.



Cartridge Filter

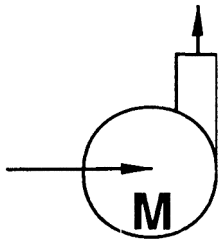
Most often used where extremely fine filtration is required. The fluid to be filtered flows across a set of **cartridges**. The cartridges normally consist of a metal or plastic support cage which is wrapped with layers of fine cloth or fibres. The filter cannot normally be backwashed or cleaned whilst it is in service and is therefore most often used as a **polishing** filter. The flanged top connection allows the filter cartridges to be replaced.

1.20 PUMPS AND COMPRESSORS



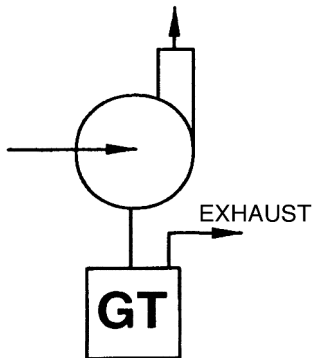
Centrifugal Pump

The most common pump used in the oil and gas industry. The pump may be used in almost any service. Capacity may range from a few cubic metres per hour to around a thousand cubic metres per hour. The centrifugal pump is used mainly in **constant pressure/variable volume** services. Extremely high pressures can be achieved by the use of multi-impeller pumps. Regardless of the size of pump, the symbol almost always remains the same.



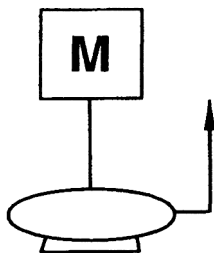
Centrifugal Pump

Alternative symbol. In this instance the pump is being driven by an electric motor as indicated by the letter "M".



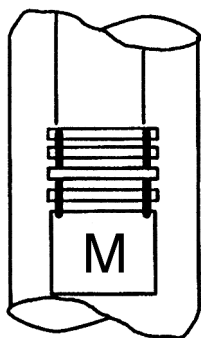
Centrifugal Pump

Alternative symbol. In this instance the pump is being driven by a gas turbine. (If the main driver is not indicated on the Piping and Instrument Diagram then almost certainly an electric motor is being used.)



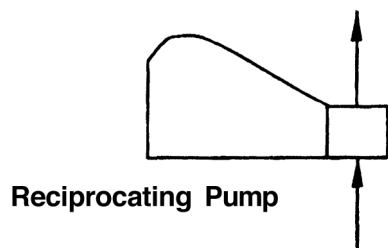
Vertical Sump Pump

A variation of the centrifugal pump. The sump pump is designed to hang vertically below the level of the liquid being pumped. Smaller models are often driven by small pneumatic (air powered) motors.

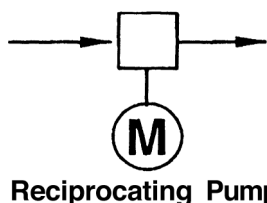


Electrical Submersible Pump (ESP)

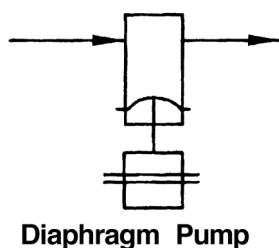
A multi-stage (often over 34 stages) centrifugal pump usually fitted to high volume wells which will not flow without mechanical assistance. Sizes may vary from as low as 200 barrels per day to over 20,000 barrels per day.



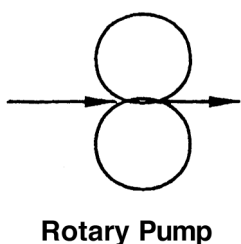
Used mainly in services which require relatively low flows at high differential pressures.



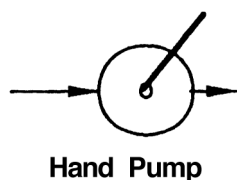
An alternative symbol. For chemical injection service it may be provided with a variable flow feature or in **multi-head** arrangements whereby six or more pumps are powered by a single motor.



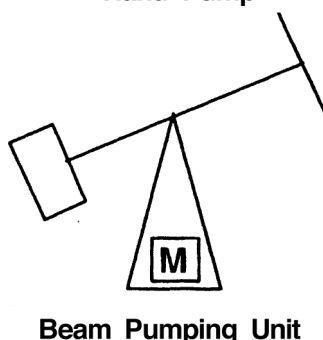
Used in low flow and relatively low differential pressure service. Excellent type of pump for dirty services.



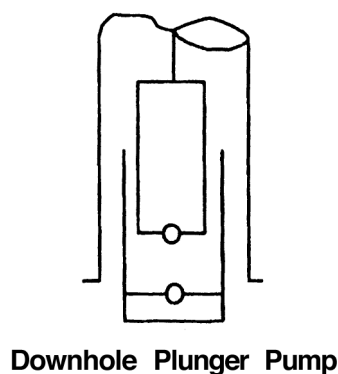
A positive displacement pump with a rotary action, (ie uses meshed gears, screws or lobes to generate the pressure and flow). Often used in services where a relatively high pressure is required, and the liquid to be pumped is clean, (eg lubricating oil, seal oil etc).



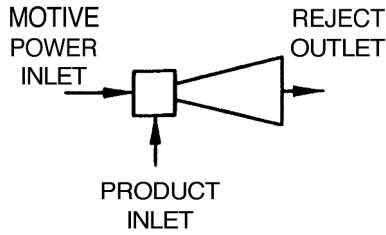
Either rotary or reciprocating action. Installed where there may be a regular requirement to unload barrels or empty small sumps.



Used exclusively onshore to pump low volume wells which will not flow without mechanical assistance.

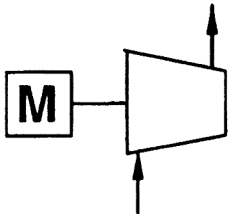


The reciprocating pump which is driven by the Beam Pumping Unit.



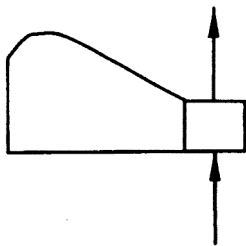
Ejector/Ejector/Venturi

Can be classed as a **pump** or as a **compressor** depending on the fluid being handled. The device is used for a variety of low pressure / high volume services. **The motive power** may be a high pressure gas or a high pressure liquid (usually air and water respectively).



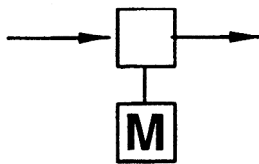
Centrifugal Compressor

The most common type of compressor found in the oil and gas industry. The compressor may be used in almost any service. Capacity may range from a few cubic metres per hour to many thousands of cubic metres per hour. The centrifugal compressor is used mainly in **constant pressure / variable volume** services. Extremely high pressures can be achieved by the use of multi-impeller compressors. Regardless of the size of compressor, the symbol almost always remains the same.



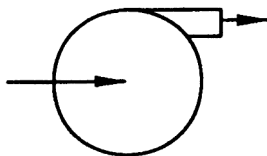
Reciprocating Compressor

Used mainly in services which require relatively low flows at high differential pressures.



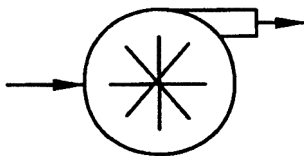
Alternative Symbol.

Reciprocating Compressor



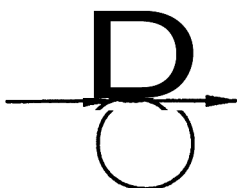
Centrifugal Blower

A low pressure / large volume compressor used for such services as ventilation or air conditioning.



Positive Displacement Blower or Vacuum Pump

Normally used in similar service to the centrifugal blower where a higher pressure is required or where the suction may be under a partial vacuum.

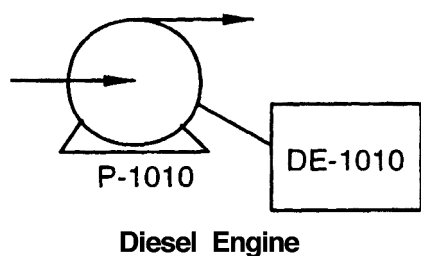


Rotary Compressor

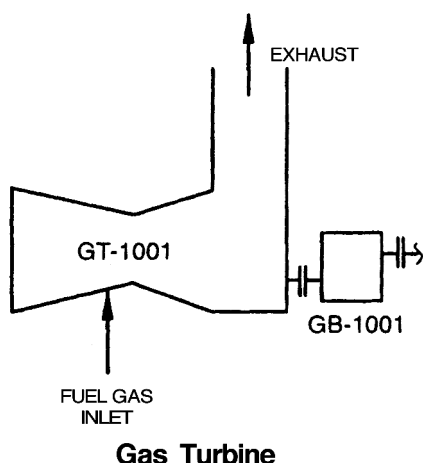
A positive displacement compressor with a rotary action, (ie uses meshed gears, screws or lobes to generate the pressure and flow). Often used in instrument air or plant air where a relatively high differential pressure is required for low flows.

It should be noted that in a number of cases the symbol for a particular pump and a particular compressor is identical, (eg reciprocating pump and reciprocating compressor.) When this occurs reference will have to be made to the identification lettering of the equipment to establish the type of equipment. As a general rule, pumps will be identified with the letter “P” and compressors will be identified with the letter “C” or “K”.

As already indicated, the type of equipment used to drive the pump or compressor may be indicated. The method of identification may be as illustrated below.

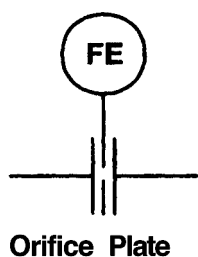


Often used to drive emergency generators, firewater pumps etc. The Piping and Instrument Diagram may also include such items of equipment as fuel tanks, exhaust spark arrestors etc.



Often used to drive power generation facilities, large capacity centrifugal pumps and compressors and occasionally emergency generators. The Piping and Instrument Diagram may also include fuel gas supply equipment, exhaust waste heat exchangers etc.

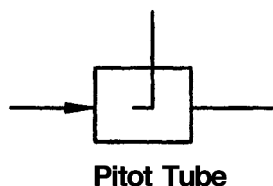
1.21 METERING DEVICES



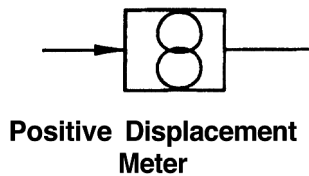
Used in both liquid and gas service. The differential pressure across the restriction is measured and used to calculate the amount of fluid flow. Light in weight, relatively cheap to produce, easy to install and maintain. The orifice plate is the most common type of metering device in use.



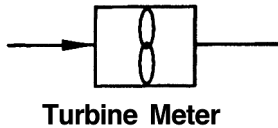
Used in both liquid and gas service. The differential pressure across the restriction is measured and used to calculate the amount of fluid flow. More accurate than the orifice plate but heavier and more expensive. Mostly used where a high pressure drop across the measuring device cannot be tolerated (eg compressor suction lines).



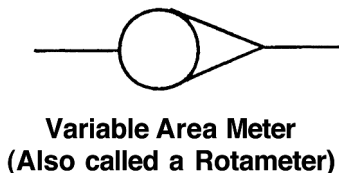
Used in both liquid and gas service. Uses the **ram** effect of the fluid hitting the end of an open pipe to generate a differential pressure which is measured and used to calculate the amount of fluid flow. Not very accurate when compared to the orifice plate and the venturi but can cope with large variations of flow. Often used in flare headers.



Used in liquid service. Extremely accurate. Individual compartments fill and empty as the liquid passes through the meter. The number of compartments filled and emptied gives an accurate measure of the liquid passing through the meter. Positive displacement meters are used on garage forecourts and are used to calibrate meter prover loops.



Used in both liquid and gas service. The fluid flow spins a turbine (ie a propeller). The number of times that the turbine rotates is an indication of the amount of fluid passing the turbine. The rotation of the turbine is measured and the fluid flow calculated from the measurement.



Used in both liquid and gas service for the accurate measurement of small flow rates. The fluid flows upwards through a conical tube. A ball or small conical weight is suspended by the flow. The flow is measured in relation to the height at which the ball or weight is suspended.

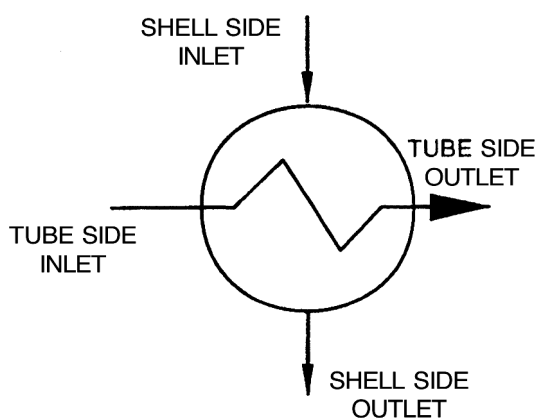
1.22 HEAT EXCHANGERS

As the name implies, a **HEAT EXCHANGER** is an item of equipment which is specifically designed to exchange heat between two substances. Heat exchangers are most often named in accordance with their function. They will be called **coolers, heaters, chillers, reboilers** etc, depending upon their function.

With shell and tube heat exchangers, the following general rules will apply :

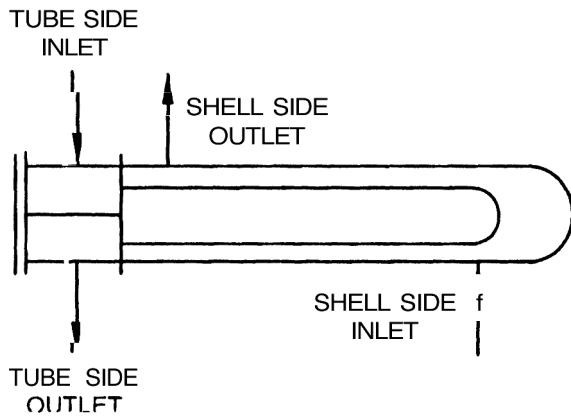
- the high pressure fluid will be routed through the tube side of the heat exchanger
- the fluid most likely to cause fouling will be routed through the tube side of the heat exchanger
- the most corrosive fluid will be routed through the tube side of the heat exchanger

In situations where both fluids fall into one or more of the above categories the designer will compromise to give the best operating results. An example of this situation is where a high pressure gas is being cooled by seawater. The gas is at high pressure and should be routed through the tube side of the heat exchanger. The seawater is corrosive and likely to cause fouling and should also be routed through the tube side of the heat exchanger. In this particular case the high pressure gas would probably be routed through the tube side of the heat exchanger.



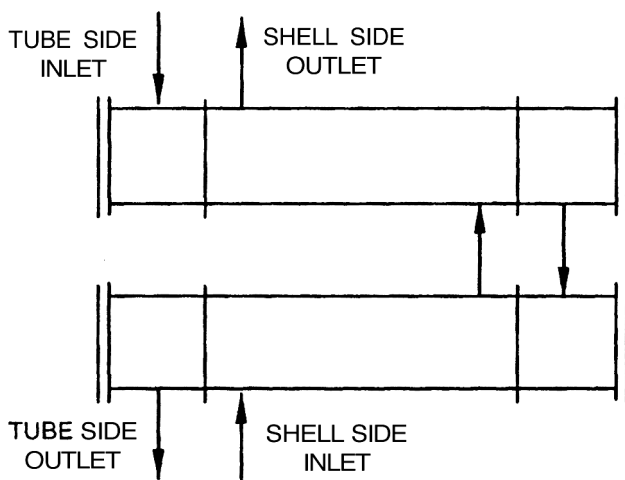
Shell and Tube Heat Exchanger

Most common type of heat exchanger. The tubes are indicated by the kinked line. The shell is indicated by the circle.



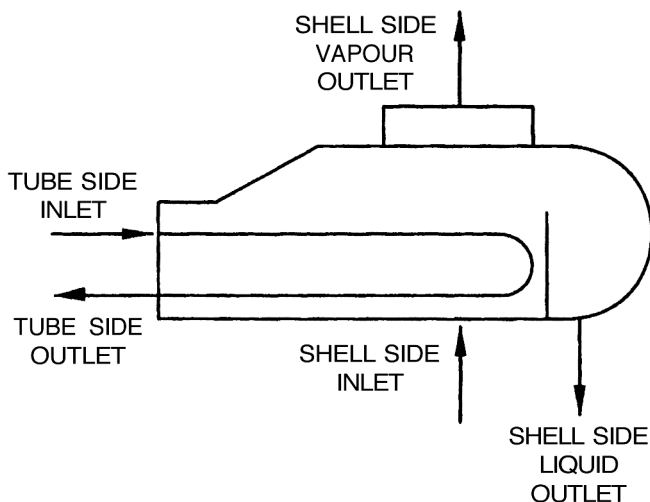
Shell and Tube Heat Exchanger

Alternative symbol. This type of exchanger may also be called a “U” Tube Heat Exchanger.



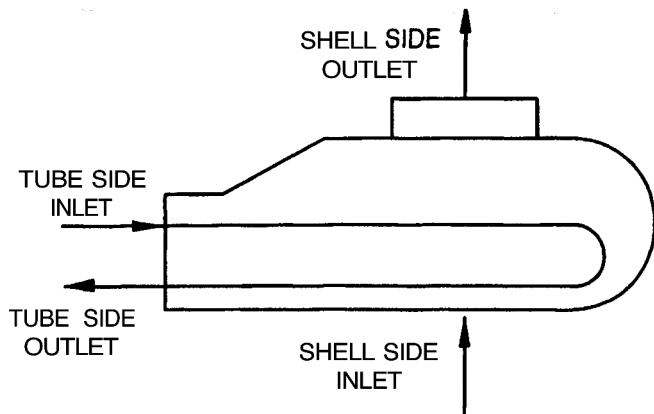
Banked Shell and Tube Heat Exchangers

Two shell and tube heat exchangers are often **banked** together. This system is often used where a single **off the shelf** heat exchanger would be too small and / or where room is at a premium.



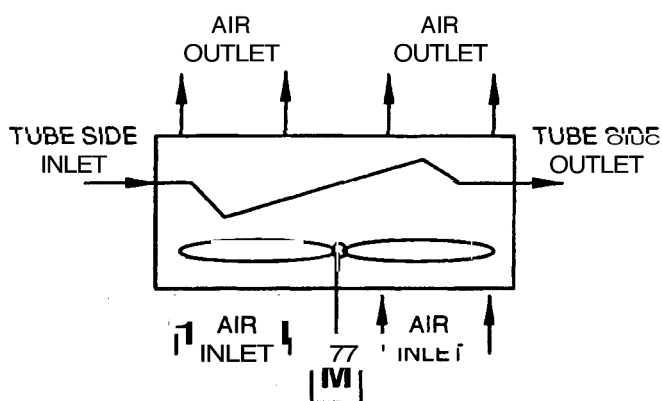
Kettle Type Heat Exchanger

With this particular kettle type heat exchanger, the shell side fluid enters as a liquid and is partially vaporised within the shell. The vapours leave the shell side from the top. The remaining liquid flows over an internal weir and leaves the shell under level control. Reboilers are normally this type of heat exchanger.



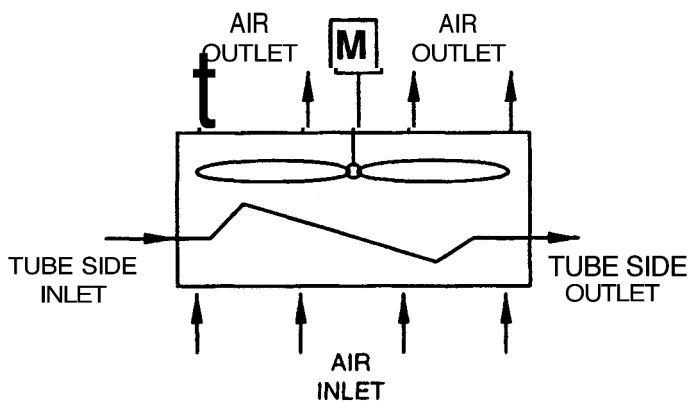
Kettle Type Heat Exchanger

With this kettle type heat exchanger, the shell side fluid enters as a liquid and is fully vaporised within the shell. The vapours leave the shell side from the top. Chillers are normally this type of heat exchanger.



Forced Draft Air Cooler
(Also called a **Fin-Fan Cooler**)

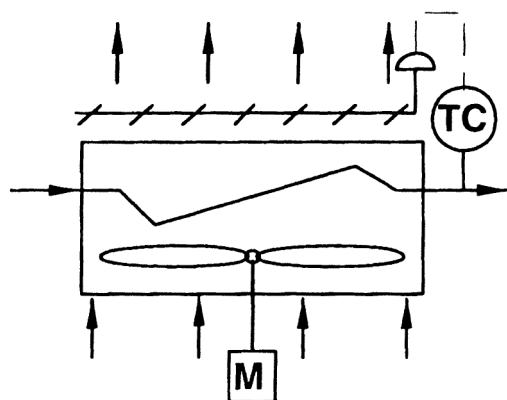
Used in a variety of services where the ambient air temperature provides sufficient cooling for the process. The tubes are normally wrapped with light alloy fins to assist in the heat transfer process. The term **forced draft** is used because the air is pushed (**forced**) across the tubes.



Induced Draft Air Cooler
(Also called a **Fin-Fan Cooler**)

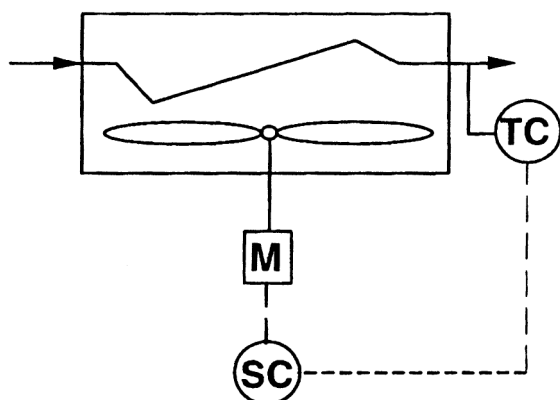
The term **induced draft** is used because the air is pulled (**induced**) across the tubes.

To control the heat exchange rate on an air cooler the flow of air across the cooler is increased or decreased. The easiest way to do this is for the Operator to switch the fan on and off. This gives very coarse control. Other control methods are illustrated below.



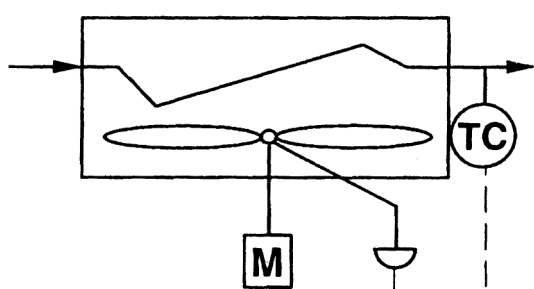
Air Cooler-Control by Louvres

A temperature controller adjusts a set of louvres to increase or reduce the amount of air flowing across the tubes.



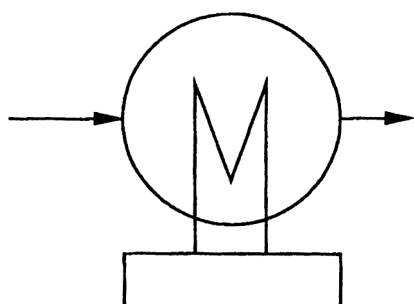
Air Cooler-Control by Fan Speed

A temperature controller adjusts the speed of the motor to increase or reduce the amount of air flowing across the tubes.



Air Cooler-Control by Fan Pitch

A temperature controller adjusts the angle of the blades on a fixed speed fan to increase or reduce the amount of air flowing across the tubes.



Electric Heater

Electric heaters may be provided where the amount of heat required is extremely low or in areas where a naked flame may be hazardous. Electric heaters are commonly controlled by a device called a **thyristor**. The thyristor is basically a very rapid switching mechanism which can adjust the amount of time that the heater is switched ON or switched OFF.

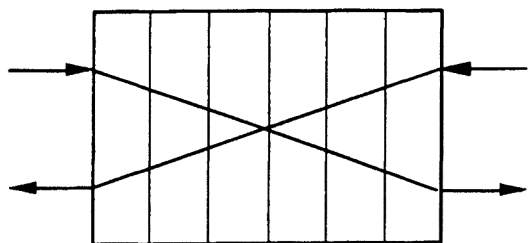
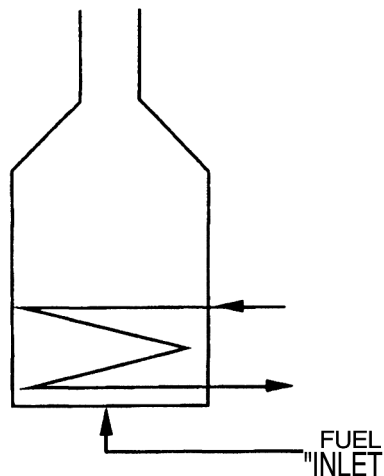


Plate Heat Exchanger

Commonly found in low pressure services. This type of heat exchanger is extremely efficient and is therefore relatively small and light for the amount of heat which can be exchanged. Because of these features the plate heat exchanger is becoming very popular offshore. It is prone to fouling but can easily be dismantled for cleaning.

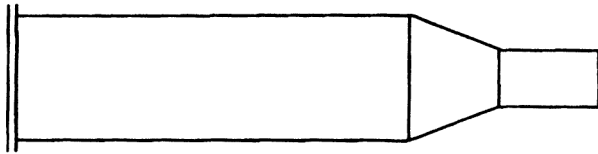


Fired Heater

Used for heavy duty service both onshore and offshore. May be used to provide heat for a wide variety of purposes from steam generation to reboiler service. The fired heater may burn a variety of fuels ranging from gas to heavy fuel oil.

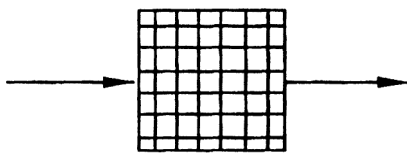
1.23 OTHER ITEMS OF EQUIPMENT

There is a wide variety of equipment which may be found on an offshore installation or an onshore facility. I have selected a few items to illustrate the variety of items which may be shown on almost any Piping and Instrument Diagram.



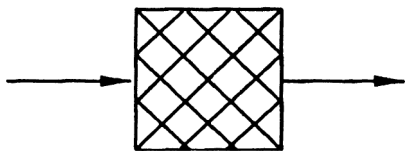
Pig Launcher/Receiver

Fitted to the ends of pipelines for the launching and recovery of the pigs or spheres used to clean the pipeline.



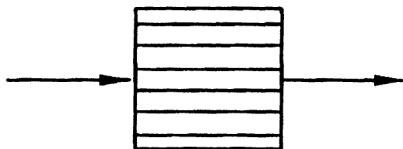
Air Filter

Fitted to the inlet of instrument air compressors, gas turbines, diesel engines or any other location where a limited amount of filtration is required prior to the air entering the process.



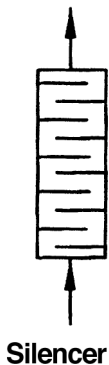
Flame Arrestor

Fitted to tank vents, low pressure flares, gas vents or any other location where a flammable atmosphere may be discharged to the air. Flame arrestors work by removing heat from the flame front to prevent the flame migrating into the pipe to which they are fitted. If the flammable atmosphere is ignited, the flame arrester prevents the flame from **back flashing** into pipes and vessels.



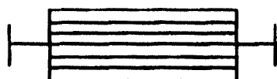
Spark Arrestor

May be an identical symbol to that used for a flame arrestor. Fitted to the exhausts of internal combustion engines such as diesel engines on emergency generators, mobile compressors etc. They prevent the emission of sparks from the exhausts and enable the engines to be used in potentially hazardous areas.



Silencer

Fitted wherever there is a need to reduce or muffle the noise of an expanding gas. They may be extremely large (eg when fitted to the exhaust of a gas turbine) or very small (eg when fitted to the exhaust of an air operated valve).



Flow Straightening Vanes

Fitted upstream of meters (especially turbine meters) to reduce the **swirling** of the fluid being measured and thereby increase the accuracy of the metering.

1.24 EQUIPMENT IDENTIFICATION

Each item of equipment, and each instrument, will have a unique identification. The identification will normally be a combination of letters and numbers. The most common uses of identification letters are shown below. As with the symbols, the identification of the equipment may differ from project to project. On one system item "E-101" may refer to **Heat Exchanger 101** on another project "E-101" may refer to **Engine 101**. I have illustrated the most popular usage of the letters.

It should be noted that:

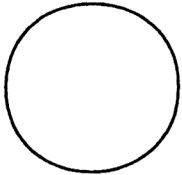
- **First** is the most popular usage and **Second** is an alternative usage.
- A **Special Item** is almost anything which does not appear regularly on the Piping and Instrument Diagram (eg an insulating gasket, a temporary strainer etc).
- In some instances two letters may be used (eg GT = Gas Turbine or DE = Diesel Engine).

LETTER	FIRST	SECOND	LETTER	FIRST	SECOND
A	ANALYSER		N		
B	BURNER		O		
C	COMPRESSOR	COLUMN	P	PUMP	
D	DRUM	DIESEL	Q	FLOW METER	
E	HEAT EXCHANGER	ENGINE	R		
F	FILTER	FLARE	S	SEPARATOR	STRAINER OR SPECIAL ITEM
G			T	TANK	TURBINE
H	HEAT EXCHANGER		U	SPECIAL ITEM	
I			V	VESSEL	VALVE
J	ELECTRICAL PANEL		W	WELL	
K	COMPRESSOR		X	HEAT EXCHANGER	COMMON
L			Y		
M	MOTOR		Z		

1.25 INSTRUMENT IDENTIFICATION

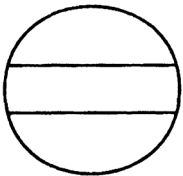
On the Piping and Instrument Diagrams the instruments are identified by their **Location**, by their **Function** and by their **Number**. The basic identification symbol for an instrument is a circle which encloses the other information.

The actual point at which the instrument is connected to the process will be indicated by a line between the circle and the process. The actual location of the instrument itself will be indicated by the design of the circle.



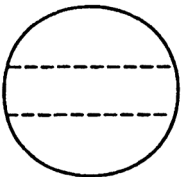
eg pressure gauge, sight glass, thermometer etc.

Mounted Locally



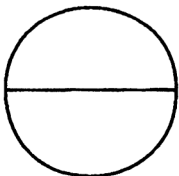
eg compressor lube oil pressure gauge, pump suction valve status light etc.

**Mounted on
Local Panel**



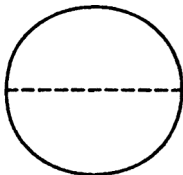
eg compressor low lube oil pressure switch, pump suction pressure switch etc.

**Mounted at rear
of Local Panel**



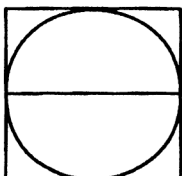
eg vessel level indicator controller, ESD valve position light etc.

**Mounted on Main
Control Room Panel**



eg compressor anti-surge by-pass switch, pump low flow shutdown by-pass switch etc.

**Mounted at rear of
Main Control Room Panel**



Contained within a Computerised System (ie a Central Computer Control and Monitoring System [CCCMS] or a Supervisory Control and Data Acquisition [SCADA] System.): eg vessel level, valve status, compressor speed etc. The information can normally only be accessed via a computer VDU screen or a print-out facility.

The **Function** of the instrument is indicated by a series of letters contained within the top half of the circle. Before we look at a few examples take the time to study the matrix laid out below. The matrix identifies the most common usage of letters which are used as **identifiers**.

LETTER	FIRST LETTER		SUCCEEDING LETTERS		
	MEASURED OR INITIATING VARIABLE	MODIFIER	READOUT OR PASSIVE FUNCTION	OUTPUT FUNCTION	MODIFIER
A	ANALYSIS		ALARM		
B	BURNER	BURSTING	BLOWDOWN	USER'S CHOICE	USER'S CHOICE
C	CONDUCTIVITY (ELECT)			CONTROL	CLOSE
D	DENSITY (MASS) OR S.G.	DIFFERENTIAL	DISC		
E	EMERGENCY	PRIMARY ELEMENT			
F	FLOWRATE	FLOWLINE (FRACTION)			
G	GAUGE (DIMENSION)		GAUGE OR GLASS		
H	HAND (MANUAL)				HIGH
I	CURRENT (ELECT)		INDICATE		
J	POWER	SCAN			
K				CONTROL STATION	
L	LEVEL		LIGHT (PILOT)		LOW
M	MOISTURE OR HUMIDITY				MIDDLE OR INTERMEDIATE
N	USER'S CHOICE		USER'S CHOICE	USER'S CHOICE	USER'S CHOICE
O	USER'S CHOICE		ORIFICE OR RESTRICTION	CUT-OUT	OPEN
P	PRESSURE OR VACUUM		POINT (TEST CONNECTION)		
Q	QUANTITY OR EVENT	INTEGRATE OR TOTALISE	INTEGRATING OR SUMMATING		
R	RELIEF		RECORD OR PRINT		RUN
S	SPEED OR FREQUENCY	SHUTDOWN SAFETY	SOLENOID OPERATED	SWITCH	STOP
T	TEMPERATURE			TRANSMITTER	
U	MULTI-VARIABLE		MULTI-FUNCTION	MULTI-FUNCTION	MULTI-FUNCTION
V	VIBRATION	VENT		VALVE, LOUVRE OR DAMPER	
W	WEIGHT OR FORCE		WELL		
X	UNCLASSIFIED		UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED
Y				RELAY OR COMPUTE	
Z	LIMIT (SWITCH)		UNCLASSIFIED	UNCLASSIFIED	UNCLASSIFIED

From the matrix you can see that:

- a First Letter “L” identifies a **Level**,
- a Second Letter “G” identifies a **Gauge**.

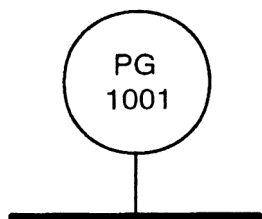
Therefore “LG” identifies a **Level Gauge**.

We can also see that:

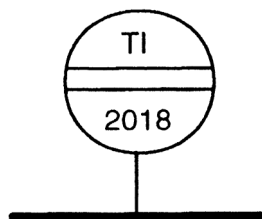
- “LALL” is a **Level Alarm Low-Low**, and
- “FIC” identifies a **Flow Indicator Controller**.

Try a few combinations yourself before moving on to the next page.

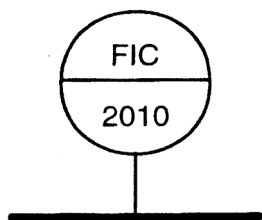
In the section on piping I indicated that there was a **System Unit Number** for each pipe. In most instances the **System Unit Number** will be carried on into the instrument identification numbering system. A few examples are indicated below :



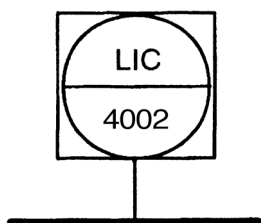
Locally mounted pressure gauge, number 01, installed in System 10.



Local control panel mounted temperature indicator, number 18, installed in System 20.



Control room panel mounted flow indicator controller, number 10, installed in System 20.



Computerised level indicator controller, number 02, installed in System 40.

Section 2 - PRACTICAL APPLICATION OF SYMBOLS

2.1 PLATE TYPE HEAT EXCHANGER

We will now look at a small section of a P&ID to see a practical application of some of the symbols. Take the time to study **Figure 18** on page 42, before carrying on.

The first thing we can see is that the Piping and Instrument Diagram is of a Plate Type Heat Exchanger ~, numbered “E-1001 A”. This tells us that it is in “System 10”. From the letter “A” we can assume that there is at least one other heat exchanger in identical service, (ie somewhere else is heat exchanger “E-1001B” and maybe “E-1001C”, “E1001D” etc.)

The piping commodity letters “PO” indicates that the Production Header, and all associated pipework, is in **Produced Oil** service. E-1001 A is supplied with heating medium from a **Heating Medium Supply Header** which indicates that the function of E-1001 A is to heat up the produced oil.

The temperature of the oil leaving E-1001 A is measured by a locally mounted temperature element (TE-1014). The electronic signal from TE-1014 goes to a control room panel mounted temperature indicator controller (TIC-1014). The temperature indicator controller is set to control the produced oil temperature at 65°C. The electronic signal leaving TIC-1014 goes to a locally mounted temperature relay (TY-1014). The temperature relay converts the electronic signal to a pneumatic signal (as indicated by the letters “I/P”). The pneumatic signal flows through a solenoid operated valve (SOV-1014) to a control valve (TV-1014) to adjust the amount of heating medium entering the heat exchanger.

The electronic signal from TE-1014 is also used to generate a temperature alarm high (TAH-1014) set at 70°C and a temperature alarm low (TAL-1014) set at 60°C.

A locally mounted temperature switch high-high (TSHH-1011) offers another level of protection against over temperature. TSHH-1011 is set at 75°C. If the produced oil temperature reaches 75°C TSHH-1011 will activate an alarm (TAHH-1011) on the control room panel and also send a signal to the Shutdown System.

Another input to the Shutdown System is generated by a locally mounted pressure differential switch low (PDSL-1027). A low differential pressure could indicate that a leak has occurred within the heat exchanger. If a low differential pressure occurs PDSL-1027 will activate an alarm (PDAL-1027) on the control room panel and also send a signal to the Shutdown System.

If the Shutdown System is activated by TSHH-1011, or by PDSL-1027, the power to SOV-1014 will be removed. SOV-1014 will fail to the vent position (as indicated by the small curved arrow). The air from TY-1014 would be isolated and the air would be vented from TV-1014. TV-1014 would fail to the closed position as indicated by the letters “FC”

Other features include:

- Spectacle blinds are fitted on the produced oil and heating medium lines to allow E-1001 A to be isolated for maintenance.
- Pressure relief valve (PSV-1073) set to relieve at 12 barg pressure. PSV-1073 protects the produced oil side of E-1001 A against over-pressure.

We have seen that TV-1014 is indicated as being a **fail closed** valve by the letters “FC”. The valve downstream of PSV-1073 is indicated as being “LO”. This indicates that the valve must be locked open to ensure that PSV-1073 can operate properly at all times. Valves may be indicated as being :

- FC = Fail Closed (when the motive power is removed the valve moves to the **closed** position)
- FO = Fail Open (when the motive power is removed the valve moves to the **open** position)
- FIS = Fail In Situ (when the motive power is removed the valve stays in the last position requested by the controller)
- LO = Locked Open (a physical barrier prevents the valve from being closed)
- LC = Locked Closed (a physical barrier prevents the valve from being opened)
- CSC = CAR Sealed Closed
- CSO = CAR Sealed Open

CAR sealed is an abbreviation of **Customs and Revenue** sealed. It comes from situations where Customs and Revenue Officers seal valves to ensure that they are not operated without authorisation. (eg whisky distilleries, bonded stores etc). The name has been adopted by the oil and gas industry to indicate those valves which should only be moved from the **sealed** position in an emergency situation or with the proper authority.

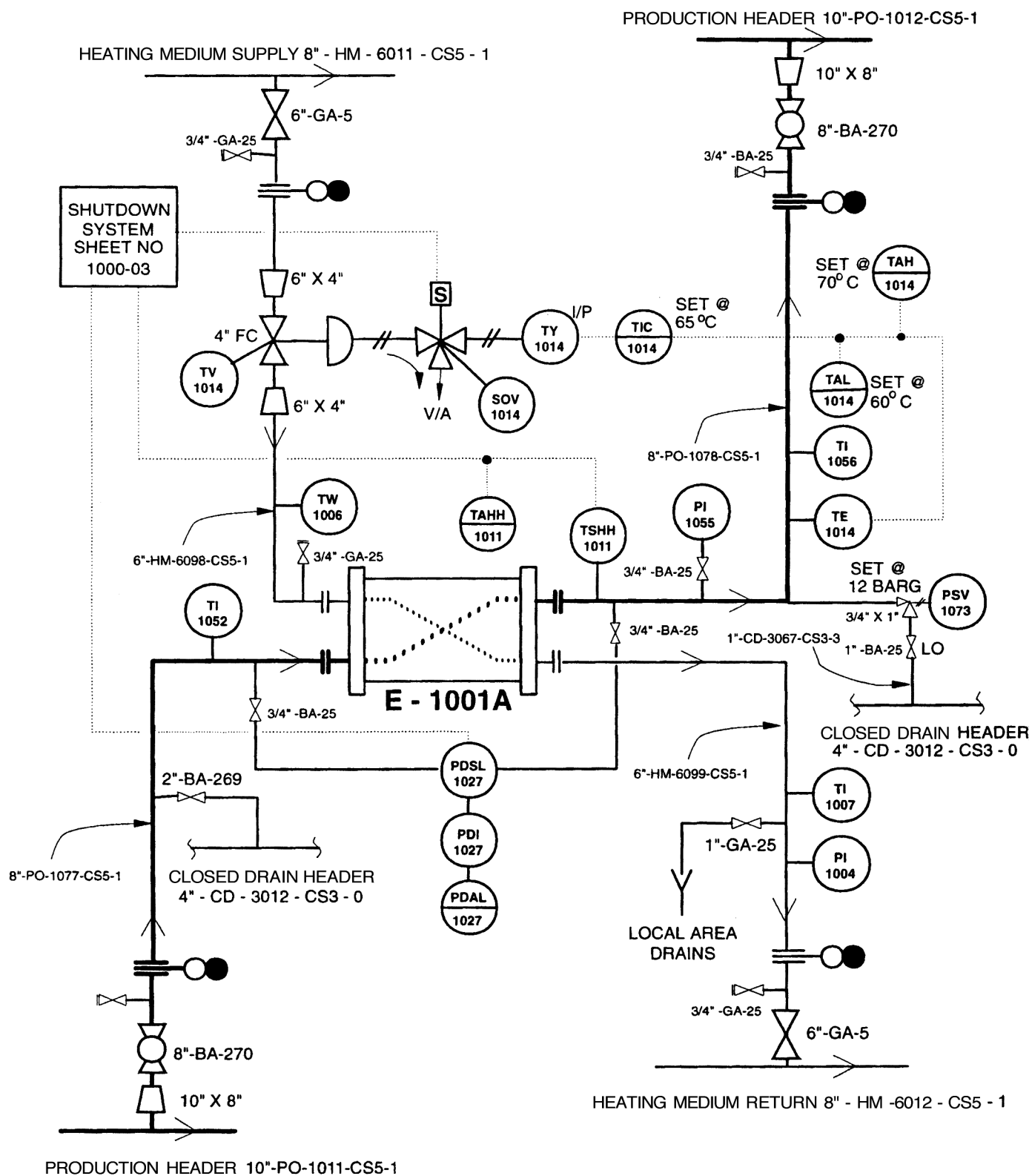


Figure 18

2.2 VALVE INTERLOCKS

Another feature which may be encountered is where two or more valves are **interlocked**. The most common example of interlocked valves is to be found where two PSV's are used to protect an item of equipment (eg a vessel or pipeline). To ensure that the equipment is protected at all times, the interlock system is designed such that at least one PSV is operational at all times. In the illustration :

- Valves A & B and Valves C & D may all be open.
- Valves A & B may be closed **only** if valves C & D are open.
- Valves C & D may be closed **only** if valves A & B are open.

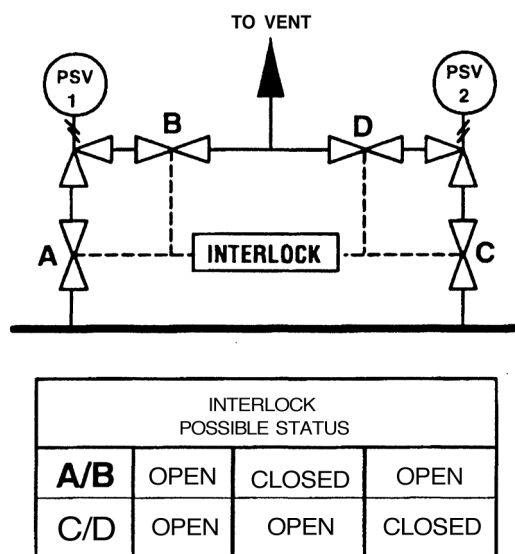


Figure 19

Section 3 - PIPING AND INSTRUMENT DIAGRAMS

Figures 7 to 17 to be used with Book 1 Sections 2, 3 and 4.

Figure 7	-	PL - 0101 - 01	DP PLATFORM - MAIN DECK LEVEL 1 - PLOT PLAN
Figure 8	-	EL - 0206 - 01	UQ PLATFORM ELEVATION - LOOKING NORTH
Figure 9	-	FD - 0002 - 01	PROCESS FLOW DIAGRAM - SEPARATION - YEAR 1
Figure 10	-	FD - 0002 - 02	PROCESS FLOW DIAGRAM - GAS TREATMENT AND COMPRESSION - YEAR 1
Figure 11	-	PD - 0002 - 01	P&ID - TOPSIDE PRODUCTION WELLHEAD TYPICAL
Figure 12	-	PD - 0016 - 01	P&ID - FIRST STAGE PRODUCTION SEPARATOR
Figure 13	-	PD - 0021 - 01	P&ID - OIL BOOSTER PUMP - P-0101A
Figure 14	-	PD - 0030 - 01	P&ID - LP COMPRESSOR SUCTION COOLER AND DRUM - TRAIN A
Figure 15	-	PD - 0031 - 01	P&ID - LP COMPRESSOR - TRAIN A
Figure 16	-	PD - 0001 - 02	GENERAL LEGEND FOR P&IDs
Figure 17	-	PD - 0001 - 01	GENERAL LEGEND FOR P&IDs

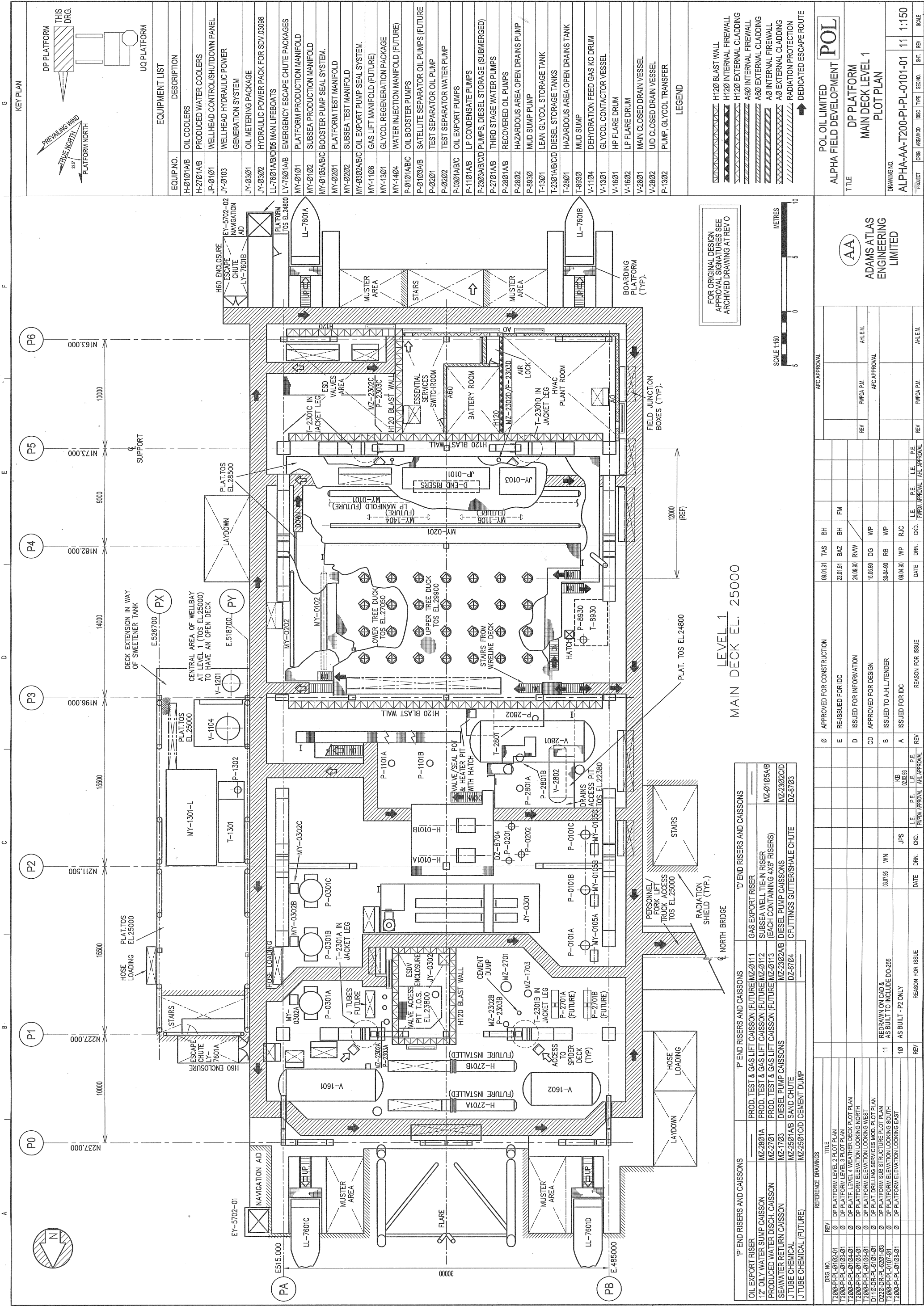


Figure 7

ZFAS/192.259P/010101.DGN

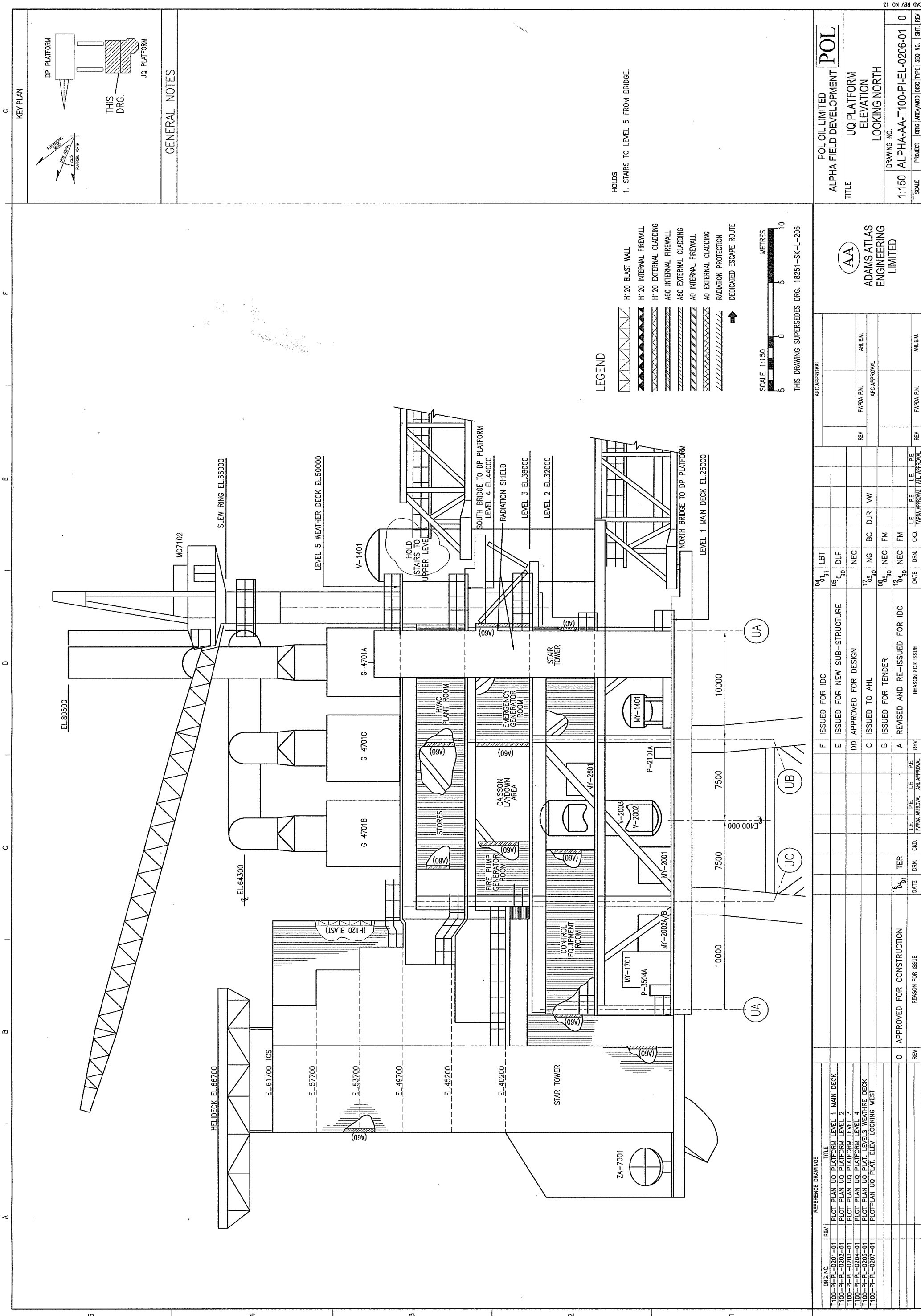


Figure 8

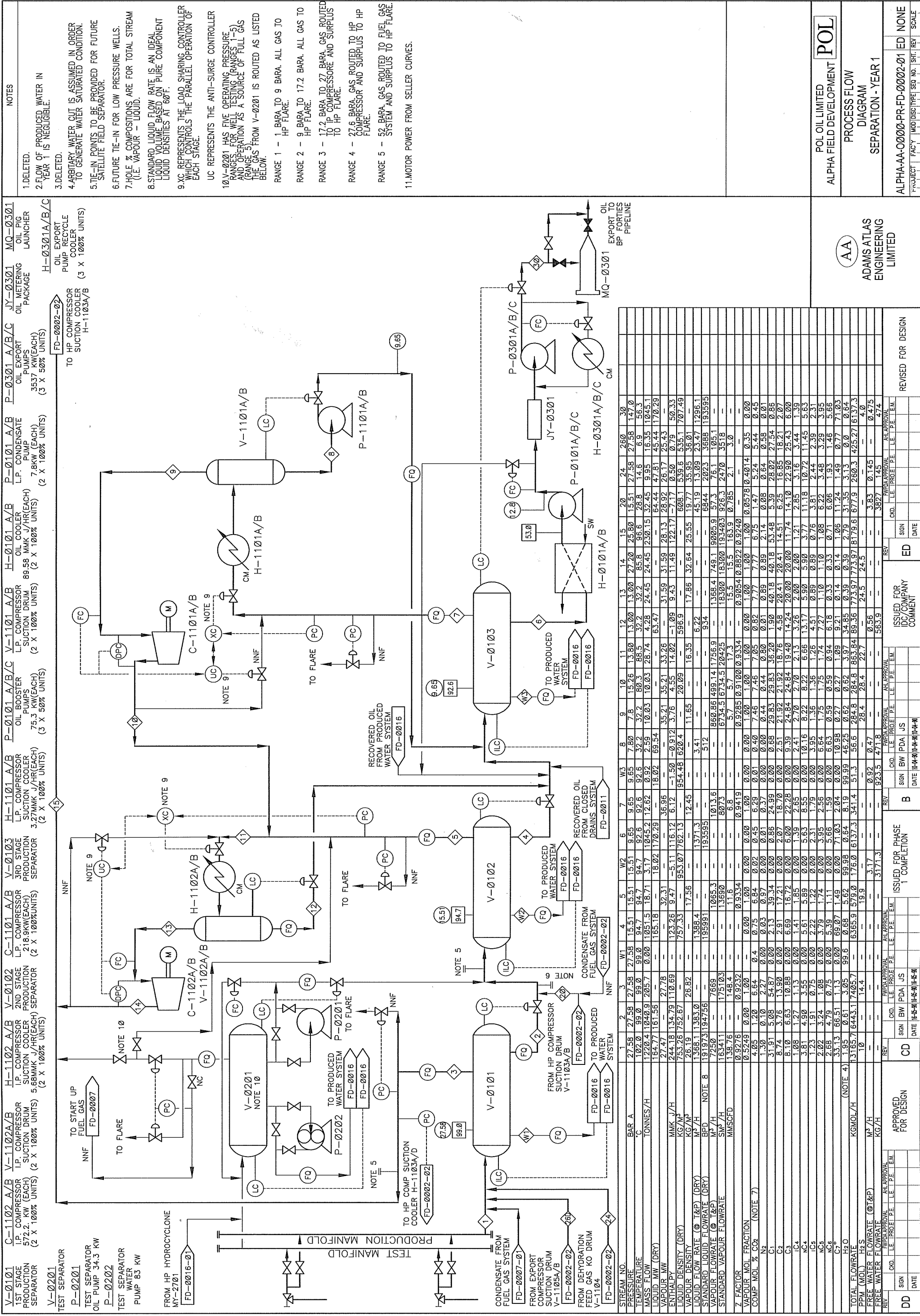
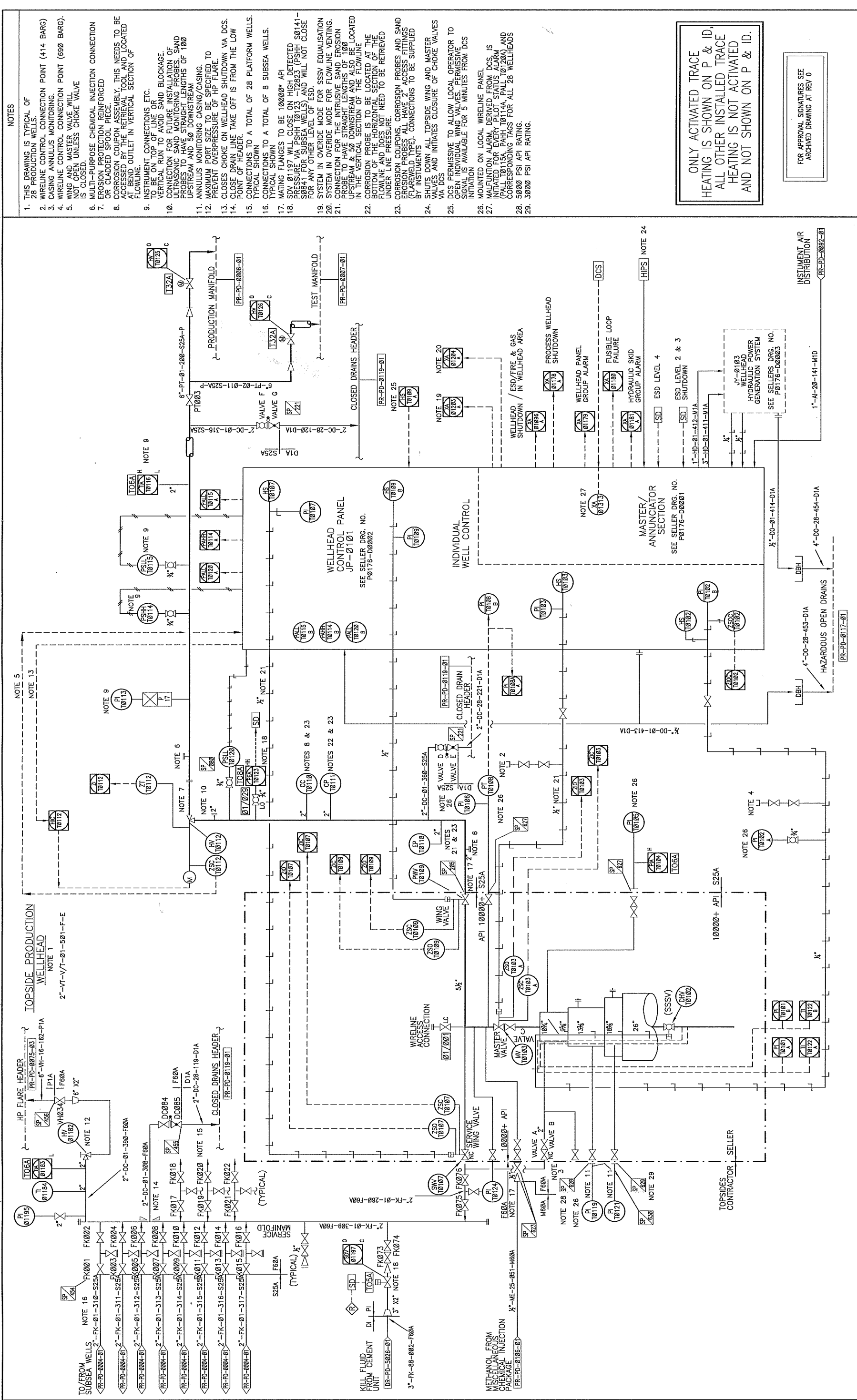


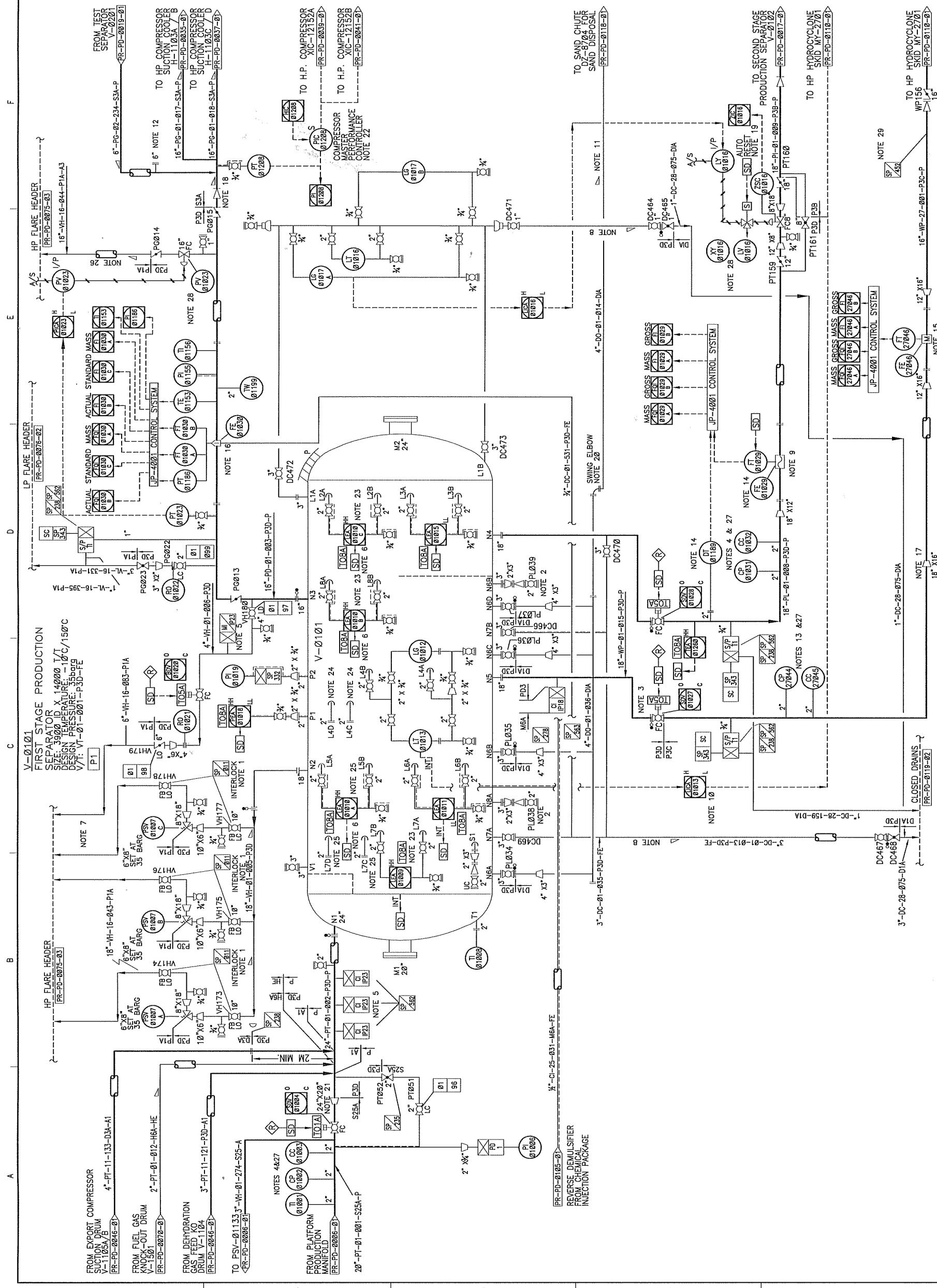
Figure 9

A B C D E F G



NOTES

1. THIS DRAWING IS TYPICAL OF 28 PRODUCTION WELLS.
2. WIRELINE CONTROL CONNECTION POINT (414 BARG).
3. CASING ANNULUS MONITORING.
4. WIRELINE CONTROL CONNECTION POINT (690 BARG).
5. WING AND MASTER VALVE WILL NOT OPEN UNLESS CHOKE VALVE IS CLOSED.
6. MULTI-PURPOSE CHEMICAL INJECTION CONNECTION OR CLADDED SPOOL PIECE.
7. EROSION PROTECTION REINFORCED.
8. CORROSION COUPON ASSEMBLY. THIS NEEDS TO BE ACCESSIBLE BY THE RETRIEVAL TOOL AND LOCATED AT BEND OUTLET IN VERTICAL SECTION OF FLOWLINE.
9. INSTRUMENT CONNECTIONS ETC.
10. TO BE ON TOP OF LINE OR VERTICAL RUN TO AVOID SAND BLOCKAGE. CONNECTION FOR FUTURE INSTALLATION OF PROBE TO HAVE STRAIGHT LENGTHS OF 100 UPSTREAM AND 50 DOWNSTREAM.
11. ANNULUS MONITORING CASING/CASING.
12. MAXIMUM PORT SIZE TO BE SPECIFIED TO PREVENT OVERPRESSURE OF HP FLARE.
13. CLOSE CHOKES ON WELLHEAD SHUTDOWN VIA DCS.
14. CLOSE DRAIN LINE TAKE OFF IS FROM THE LOW POINT OF HEADER.
15. CONNECTIONS TO A TOTAL OF 28 PLATFORM WELLS.
16. TYPICAL SHOWN.
17. TYPICAL SHOWN.
18. MATING FLANGES TO BE 100000* API. SDV 01197 WILL CLOSE ON HIGH DETECTED PRESSURE VIA PSH 10123 - 10203 (ESHH S0141-50841 FOR SUBSEA WELLS) AND WILL NOT CLOSE FOR ANY OTHER LEVEL OF ESD.
19. SYSTEM IN OVERIDE MODE FOR SSSV EQUALISATION.
20. CONNECTION FOR THE INTRUSIVE SAND EROSION PROBE TO HAVE STRAIGHT LENGTHS OF 100 UPSTREAM & 50 DOWNSTREAM AND ALSO BE LOCATED IN THE VERTICAL SECTION OF THE FLOWLINE.
21. CORROSION PROBE IS TO BE LOCATED AT THE BOTTOM OF THE HORIZONTAL SECTION OF THE FLOWLINE AND DOES NOT NEED TO BE RETRIEVED UNDER LINE PRESSURE.
22. CORROSION COUPONS, CORROSION PROBES AND SAND PROBE PROBES ALL HAVE 2" ACCESS FITTINGS (F60123, F60124, F60125, F60126, F60127, F60128, F60129, F60130, F60131, F60132, F60133, F60134, F60135, F60136, F60137, F60138, F60139, F60140, F60141, F60142, F60143, F60144, F60145, F60146, F60147, F60148, F60149, F60150, F60151, F60152, F60153, F60154, F60155, F60156, F60157, F60158, F60159, F60160, F60161, F60162, F60163, F60164, F60165, F60166, F60167, F60168, F60169, F60170, F60171, F60172, F60173, F60174, F60175, F60176, F60177, F60178, F60179, F60180, F60181, F60182, F60183, F60184, F60185, F60186, F60187, F60188, F60189, F60190, F60191, F60192, F60193, F60194, F60195, F60196, F60197, F60198, F60199, F60200, F60201, F60202, F60203, F60204, F60205, F60206, F60207, F60208, F60209, F60210, F60211, F60212, F60213, F60214, F60215, F60216, F60217, F60218, F60219, F60220, F60221, F60222, F60223, F60224, F60225, F60226, F60227, F60228, F60229, F60230, F60231, F60232, F60233, F60234, F60235, F60236, 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1. INTERLOCK SPECIFIED FOR TWO VALVES ON LINE AND ONE VALVE STAND-BY.
2. LOCATE NOZZLES FOR INTERNAL DESANDING SYSTEM.
3. LOCATE SOV 01027 AS CLOSE AS POSSIBLE TO V-0101.
4. ACCESS FITTINGS CONNECTIONS TO BE LOCATED AT THE TOP OF THE PIPE AND THE COUPON TO BE RETRACTABLE UNDER LINE PRESSURE.
5. FUTURE CHEMICAL INJECTION POINTS
6. TWO FROM THREE VOTING.
7. PSY DISCHARGES TO SLOPE TOWARDS FLARE HEADER WITH NO POCKETS. MAXIMUM SLOPE 1:50.
8. DRAIN LINES TO SLOPE TOWARDS COLLECTION HEADER. MINIMUM SLOPE 1:100.
9. MAXIMISE HEAD AVAILABLE FOR METERING PURPOSES. PROVIDE MINIMUM STRAIGHT LENGTHS 100 U/3.50 DVS
10. CONTROL TO BE STOPPED DURING EARLY PRODUCTION TO ALLOW FOR FRESH WATER PRODUCTION IS SUFFICIENT TO ENABLE MODULATING CONTROL OF LV01013.
11. SAND DISPOSAL LINES TO SLOPE TOWARDS DISPOSAL SYSTEM WITH NO POCKETS. MAXIMISE SLOPE WHERE POSSIBLE.
12. TIE-IN FOR FUTURE SATELLITE FIELD SEPARATOR.
13. CORROSION PROBES AND COUPONS IN PRODUCED WATER SERVICE TO BE LOCATED TO GIVE ADEQUATE ACCESS IN ORDER THAT THEY MAY BE RETRACTABLE UNDER LINE PRESSURE.
14. FDE-01029 AND DT-01189 ARE SUPPLIED BY JP-4001 SELLER. DT-01189 LINE CONNECTION IS VIA A 2" ANSI CLASS 600 OUT OF SPEC FLANGE.
15. FE-27046 IS SUPPLIED BY JP-4001 SELLER. 50 U/20 DVS. MIN STRAIGHT LENGTHS 100 U/3.50 DVS.
16. COMPLETE OFFICE WATER-RUN ASSEMBLY WITH UPSTREAM AND DOWNSTREAM FLOW CONDITIONING PIPE RUN SECTIONS AND PRE-INSTALLED INSTRUMENTS BY JP-4001 SELLER.
17. LOCATE REDUCER A MINIMUM OF 5M FROM V-0101. NOZZLE N5.
18. LOCATE CHECK VALVE AT HIGH POINT TO ENABLE FALL INTO H-1103A/B.
19. AUTO RESET SOLENOID VALVE.
20. SAND DISPOSAL VIA HOSE CONNECTION TO TRANSPORTABLE TANKS.
21. LOCATE REDUCER A MINIMUM OF 5M FROM V0101. NOZZLE N1.
22. BY COMPRESSOR SELLER.
23. LEA-01010 A/B/C AND LEA-01009 ARE ANALOGUE TRANSMITTERS.
24. NOZZLES L4C AND L4D ARE FOR RELOCATION OF LT-01013 AND LG-01012 IN CONJUNCTION WITH THEIR ADJUSTMENT.
25. NOZZLES L7C AND L7D ARE FOR RELOCATION OF LEA-01009 IN CONJUNCTION WITH THEIR ADJUSTMENT.
26. ACOUSTICS INSULATION TO HP FLARE HEADER.
27. CORROSION COUPONS AND CORROSION PROBES ALL HAVE 2" ACCESS FITTINGS (FLAREWELD TYPE) CONNECTIONS TO BE SUPPLIED BY INSTRUMENTS.
28. MILLI-AMPERE INDICATOR TO BE INSTALLED ADJACENT TO VALVE.
29. 90° ELBOW OUT OF SPEC.

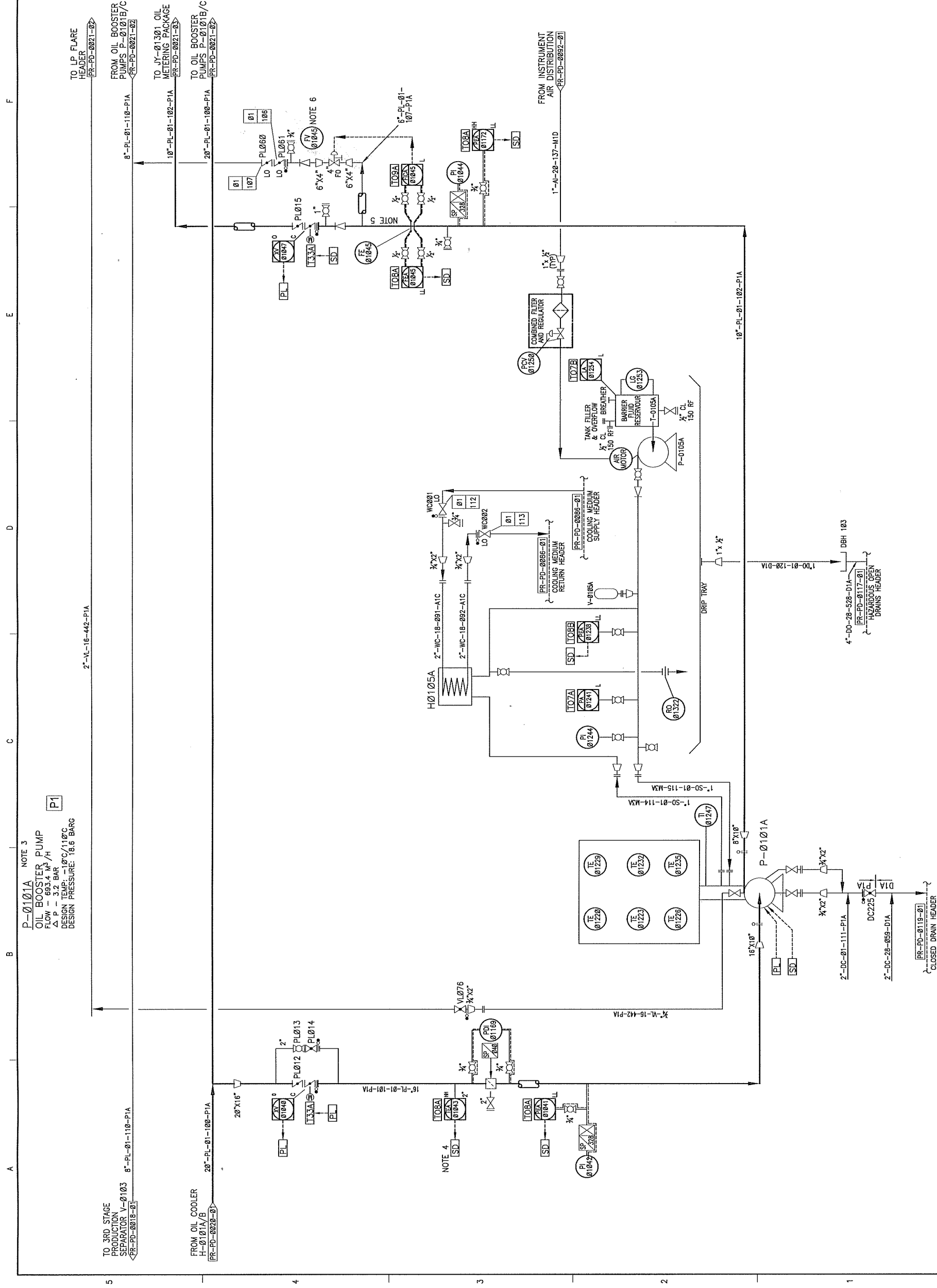
ONLY ACTIVATED TRACE
HEATING IS SHOWN ON P & ID,
ALL OTHER INSTALLED TRACE
HEATING IS NOT ACTIVATED
AND NOT SHOWN ON P & ID.

FOR APPROVAL SIGNATURES SEE
LOCAL OFFICE

[illegible]

HESSAO.REF

Figure 12

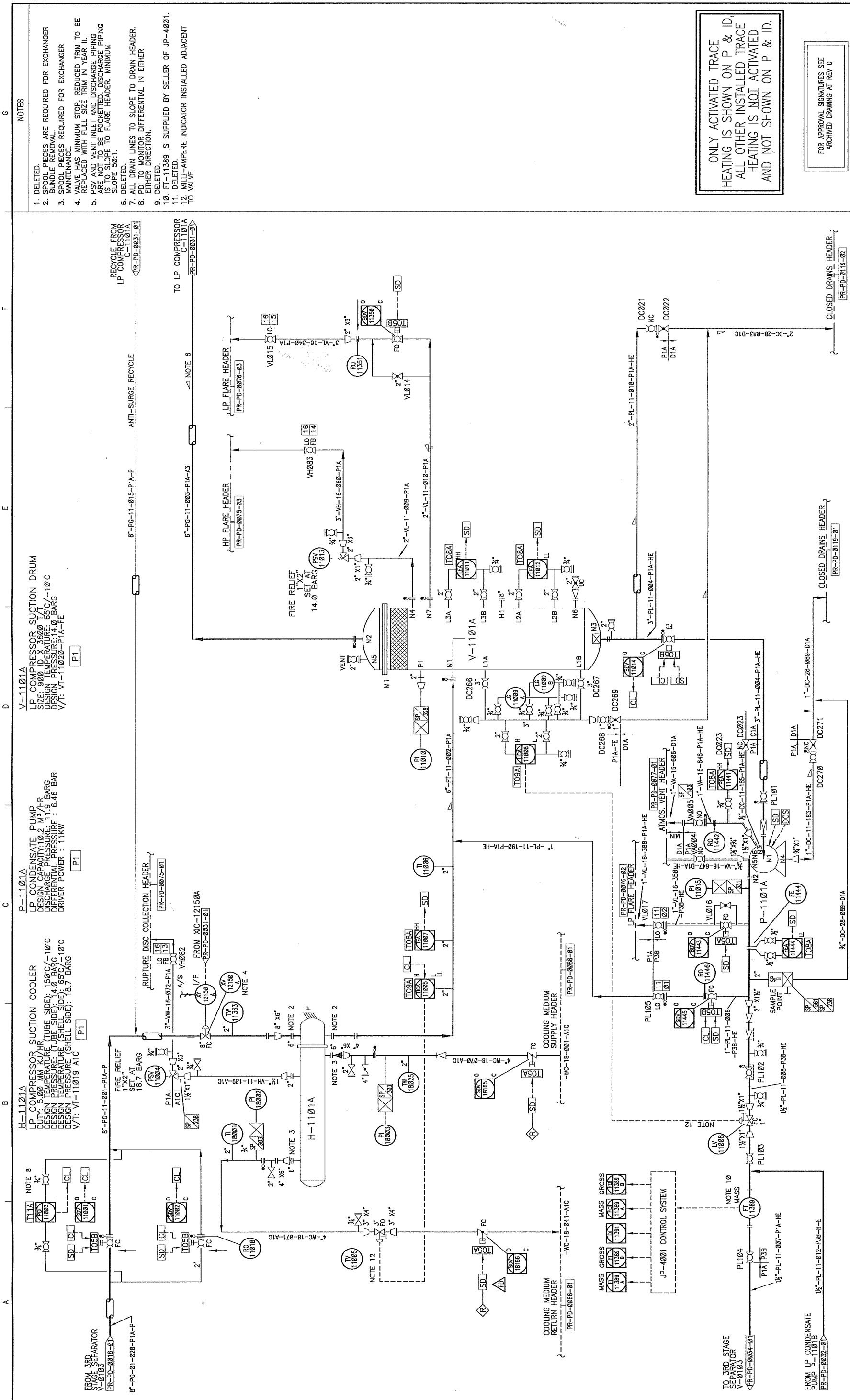


ONLY ACTIVATED TRACE
HEATING IS SHOWN ON P & ID,
ALL OTHER INSTALLED TRACE
HEATING IS NOT ACTIVATED
AND NOT SHOWN ON P & ID.

FOR APPROVAL SIGNATURES SEE
ARCHIVED DRAWING AT REV 0

[illegible]

Figure 13



REFERENCE DRAWINGS										POL OIL LIMITED ALPHA FIELD DEVELOPMENT										POL				
DRG NO.		REV		TITLE										ALPHA FIELD DEVELOPMENT										
				DCP 50000-744 ARN 3916 (EP-239)																				
				DCP 50000-286 ARN 2041 (EP-193)																				
				DCP 50000-729 ARN 3847 (EPL 128)																				
				DCP 50000-775 ARN 6328 (EP-238)																				
				DCP 50000-AH1054																				

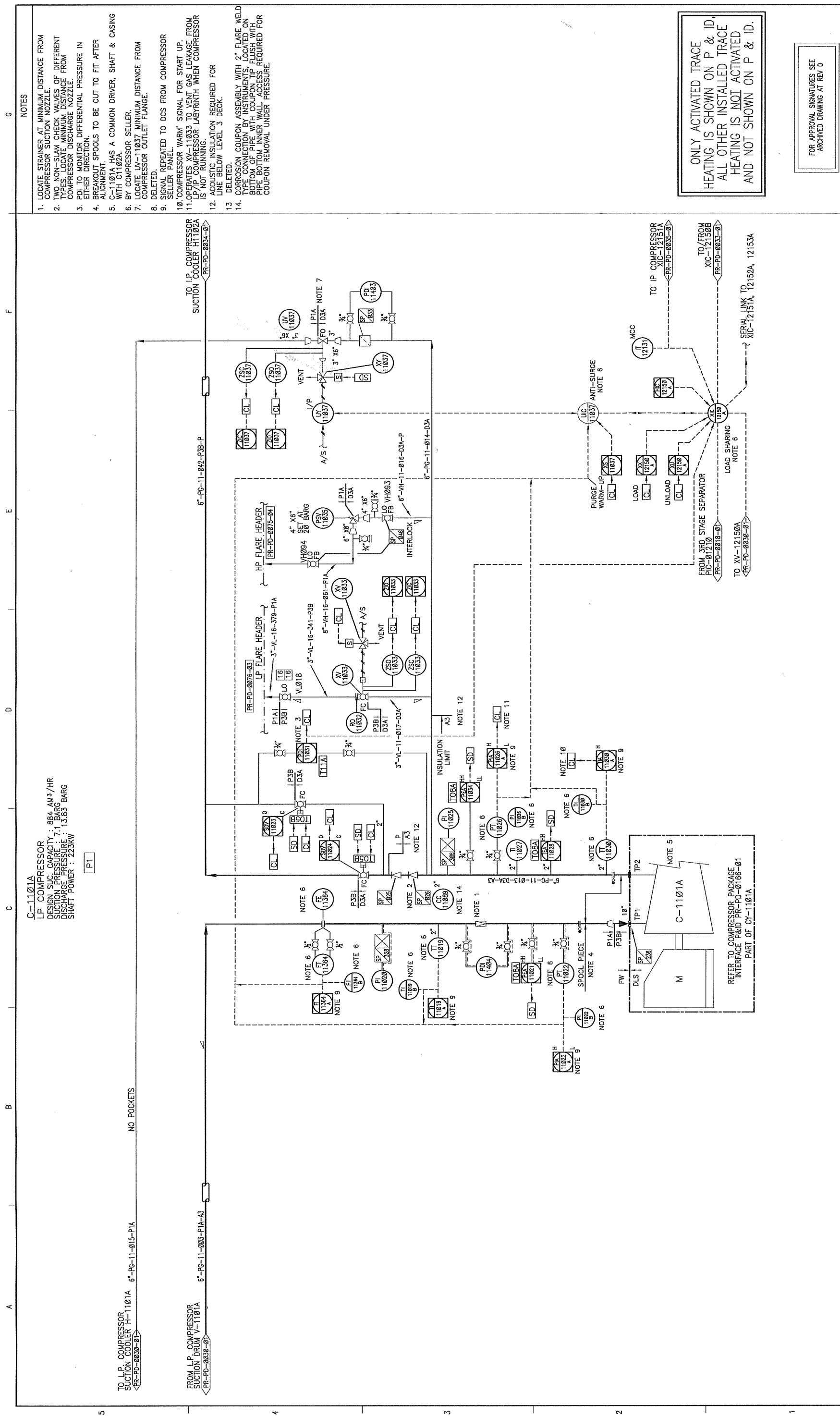
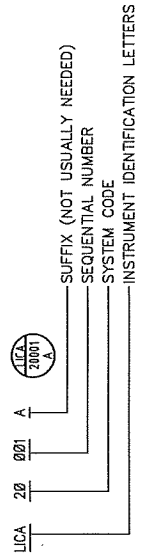
[illegible]

Figure 15

INSTRUMENT IDENTIFICATION NUMBER



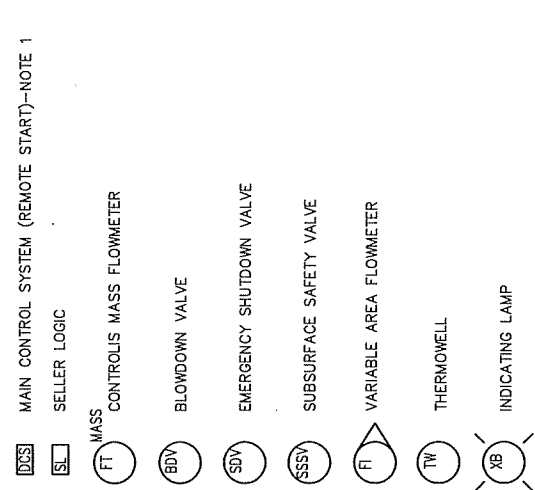
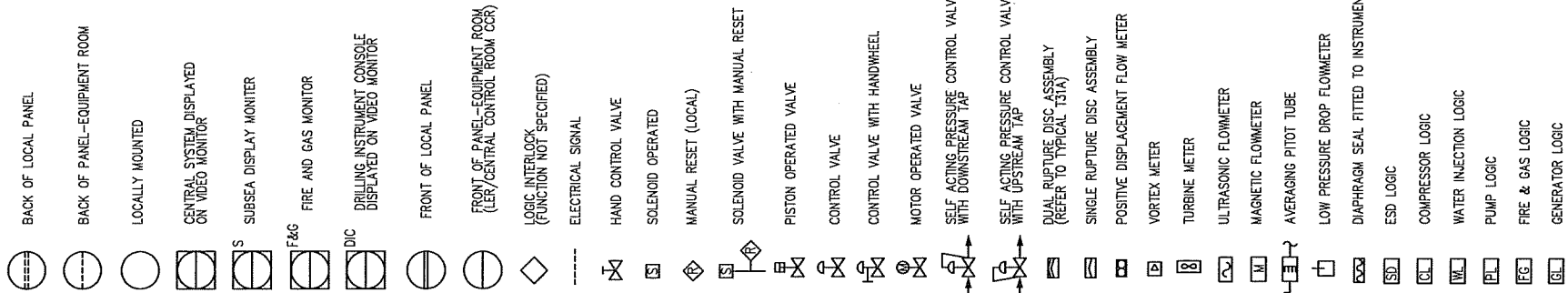
MEANINGS OF INSTRUMENT IDENTIFICATION LETTERS

First Letter	Modifier	Succeeding Letters	Modifier
A	Analysis of Measured or Counting Variable	Pressure or Function	Output Function
B	Analysis	Alarm	Modifier
C	Analysis	Users Choice	Users Choice
D	Analysis	Users Choice	Users Choice
E	Analysis	Users Choice	Users Choice
F	Analysis	Users Choice	Users Choice
G	Analysis	Users Choice	Users Choice
H	Analysis	Users Choice	Users Choice
I	Analysis	Users Choice	Users Choice
J	Analysis	Users Choice	Users Choice
K	Analysis	Users Choice	Users Choice
L	Analysis	Users Choice	Users Choice
M	Analysis	Users Choice	Users Choice
N	Analysis	Users Choice	Users Choice
O	Analysis	Users Choice	Users Choice
P	Analysis	Users Choice	Users Choice
Q	Analysis	Users Choice	Users Choice
R	Analysis	Users Choice	Users Choice
S	Analysis	Users Choice	Users Choice
T	Analysis	Users Choice	Users Choice
U	Analysis	Users Choice	Users Choice
V	Analysis	Users Choice	Users Choice
W	Analysis	Users Choice	Users Choice
X	Analysis	Users Choice	Users Choice
Y	Analysis	Users Choice	Users Choice
Z	Analysis	Users Choice	Users Choice

SPECIAL NOTE

SDV DENOTES SHUTDOWN VALVE
BDV DENOTES BLOWDOWN VALVE
DHW DENOTES DOWNHOLE VALVE
PWV DENOTES PRODUCTION WING VALVE
SWV DENOTES SERVICE WING VALVE
MV DENOTES MASTER VALVE

INSTRUMENT SYMBOLS



DRILLING SYSTEM PANEL ABBREVIATIONS

BOP 1	BLOW OUT PREVENTER PANEL (DRILL FLOOR)
BOP 2	BLOW OUT PREVENTER PANEL (REMOTE)
BOP 3	PANEL INDICATOR TOOL PUSHERS OFFICE
BSP	MAIN BULK STORAGE PANEL
BP	BARYTE PANEL
CCC	CHOICE CONTROL CONSOLE
CPI	CEMENT PANEL
DAP	DRILLERS ALARM PANEL
DFGP	DRILLERS REPEAT FIRE & GAS PANEL
DIC	DRILLERS INSTRUMENT CONSOLE
DCC	DRILLERS CONTROL CABIN
DR	DRILLERS RECORDER
MMP	MUD MIX PANEL
RCP	RAT HOLE CONTROL PANEL
VDP	VEE DOOR CONTROL PANEL
RR	REMOTE RECORDER—TOOL PUSHERS OFFICE
MCC	MOTOR CONTROL CENTRE

INSTRUMENT LINE

PNEUMATIC LINE

ELECTRICAL LINE

CAPILLARY TUBING

HYDRAULIC LINE

[illegible]

HESSA0.REF

Figure 16

